Dual-Graph Knowledge Distillation Meets Multiscale CNN: A Novel Ensemble Approach for Few-Shot Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification aims at categorizing each pixel in an HSI, facilitating precise identification and differentiation of various land cover types. In recent years, graph neural networks (GNNs)-based methods for HSI classification have exhibited impressive performance when abundant labeled samples are available. However, their effectiveness is notably constrained in real-world scenarios with limited labeled samples. Furthermore, current approaches predominantly concentrate on extracting global features at the superpixel level, overlooking the efficient exploration of pixel-level local features, leading to incomplete feature representations. To address these challenges, we introduce a novel ensemble approach termed dual-graph knowledge distillation integrated with multiscale convolutional neural network (KDMSC), specifically designed for few-shot HSI classification. Specifically, at the superpixel level, a graph structure is initially constructed for segmented superpixels, enhancing two distinct graph views. To amalgamate the dual-graph model, a dual-student graph knowledge distillation (DSGKD) module is introduced. This module intelligently merges the dual-graph structures based on superpixels, creating an ensemble teacher model that utilizes superpixel predictions as supplementary supervisory signals, thereby bolstering the model's feature learning capabilities. At the pixel level, a multiscale convolutional neural network (CNN) module is integrated to capture local spatial and spectral information across various receptive fields. This effectively explores pixel-level local features and enriches the pixel-level feature representation. Finally, an adaptive weighted fusion strategy is proposed, dynamically adjusting the contribution of global and local features. This enables the effective fusion of superpixel-level and pixel-level features, significantly enhancing the completeness and discriminability of the feature representations. Moreover, ensemble learning is introduced to further enhance the method's robustness. Experimental results on four benchmark datasets, each with only five labeled samples per class, demonstrate that the proposed method surpasses existing deep learning techniques in terms of classification accuracy and generalization performance. The source code is available online at: https://github.com/AXTX725/KDMSC**

***Index Terms*—Convolutional neural network (CNN), graph knowledge distillation, graph neural networks (GNNs), hyperspectral image (HSI) classification, limited labeled samples.**

I. Introduction

HYPERSPECTRAL image (HSI) is a unique form of imaging composed of hundreds of continuous narrowband spectral bands, spanning from the visible spectrum to the shortwave infrared and even thermal infrared bands. With its high spectral resolution and rich spatial information, HSI can accurately capture subtle differences between surface materials, making it widely applicable in fields such as mineral exploration [1], precision agriculture [2], environmental monitoring [3], and military target detection [4]. In HSI-related research, HSI classification is a crucial task, with the primary objective of categorizing each pixel into predefined semantic classes based on its spectral characteristics [5], thereby enabling accurate identification and differentiation of various land cover types.

In current deep learning techniques, the convolutional neural network (CNN) is the most widely used framework in HSI classification [6], [7], [8], [9], [10]. However, traditional convolution is a local operator that struggles to effectively model long-range dependencies between distant positions. To address this issue, attention mechanisms have been introduced to capture nonlocal context [11], [12], [13], [14], [15], [16], thereby enhancing feature extraction capabilities. Nevertheless, these mechanisms require calculating the similarity between each pixel and all other positions, resulting in high computational costs. Moreover, CNNs are designed to process grid-structured data with Euclidean properties, whereas objects in HSIs often exhibit non-Euclidean characteristics. In response, many graph neural networks (GNNs)-based methods [17], [18], [19], [20], [21] have been proposed in recent years, where objects are modeled as graph nodes, and the graph topology is leveraged to capture the non-Euclidean relationships between objects, thereby better representing the features of HSI.

Received 29 July 2025; revised 6 September 2025; accepted 4 October 2025. Date of publication 30 October 2025; date of current version 17 November 2025. This work was supported in part by the Natural Science Foundation of China under Grant 62566059, Grant 62567007, and Grant 62441701; in part by the Industrial Support Project of Gansu Colleges, China, under Grant CYZC11; and in part by the Natural Science Foundation of Gansu Province under Grant 23JRRA683. *(Corresponding author: Huifang Ma.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TGRS.2025.3627204, provided by the authors.

Digital Object Identifier 10.1109/TGRS.2025.3627204

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Recent advancements have extended graph-based learning methods from traditional supervised classification to unsupervised clustering for HSI. The S^2LGCC method [22] introduces an unsupervised self-correlated learning strategy with a locality-preserving graph convolutional embedding framework, which is designed for large-scale HSI clustering by capturing both local spectral–spatial structures and nonlocal relationships through a smoothly enhanced module and self-training strategy. Building on this, the MARP framework [23] proposes a multiadaptive receptive field mechanism for HSI classification, which employs graph attention and LSTM to automatically adjust aggregation paths, thereby reducing oversmoothing and improving robustness to noise. Further advancing unsupervised clustering, the SLCGC method [24] presents a lightweight self-supervised low-pass contrastive graph clustering network, where spectrally consistent regions and a low-pass graph denoising mechanism are utilized to enhance representation learning and suppress high-frequency noise. Finally, the AHSGC method [25] incorporates adaptive homophily structure learning with an adaptive filter graph encoder and dynamic graph update strategy, enabling improved clustering performance through iterative refinement and structure-aware learning. These studies demonstrate the potential of GNN-based graph learning methods in HSI clustering and classification tasks, especially in handling complex datasets. However, existing GNN methods still face performance limitations in HSI classification with scarce samples, and the utilization of unlabeled samples needs further improvement.

To address the aforementioned challenges, researchers have continuously explored strategies, such as transfer learning [26], [27] and few-shot learning (FSL) [28], [29], to enable models to accurately classify new classes with a limited number of labeled samples, significantly reducing the reliance on large-scale labeled datasets. However, existing work still has limited utilization of unlabeled samples, with most efforts focusing on the superpixel level and overlooking pixel-level information, thereby constraining further improvements in classification performance.

To mitigate the aforementioned challenges, this article introduces a novel ensemble approach, dual-graph knowledge distillation integrated with multiscale CNN (KDMSC), tailored for HSI classification. This approach is designed to deliver an efficient and highly generalizable model. Specifically, from the superpixel perspective, we first perform superpixel segmentation on the original HSI and then construct a graph based on the spatial adjacency relationships of the superpixels. Graph augmentation is applied to generate two distinct graph structures, enriching the feature representation and improving classification performance under FSL conditions. Next, we introduce dual-student graph knowledge distillation [30], treating the dual-graph model as peer student models and utilizing a gating mechanism [31] for their adaptive integration to build a more robust teacher model. In this process, unlabeled samples are fully utilized, enhancing the interaction learning between the student and teacher models through predictions on unlabeled samples. From the pixel perspective, to effectively reduce the noise interference of spectral information in HSI, we first apply 1-D convolution to the original HSI to suppress noise and enhance feature expression. Subsequently, to deeply explore the spectral and spatial information in HSI, a cross-scale convolution strategy is introduced to capture fine-grained spectral characteristics and local spatial correlations at different scales, enabling more accurate and comprehensive pixel-level feature extraction. Finally, to achieve effective fusion of multiview features, an adaptive weighted fusion strategy is designed. This strategy dynamically adjusts the weights of each view's features to maximize the retention of each feature's uniqueness and discriminative power, while minimizing redundant information interference between different views. This results in an organic combination of global information and local details during the feature fusion process, providing more precise and robust feature representations for subsequent tasks.

The key contributions of this article can be outlined as follows.

1) We highlight the critical significance of the graph knowledge distillation framework in HSI classification within FSL scenarios. This framework notably enhances model performance and generalization capabilities. By employing a gate mechanism for the adaptive integration of the dual-graph model, the interaction learning between the student and teacher models is reinforced, leading to a substantial enhancement in classification accuracy. Moreover, the inclusion of ensemble learning mitigates the influence of hyperparameters.
2) We introduce a multiscale CNN branch tailored to extract multiscale information from pixel-level features through cross paths. Furthermore, to optimize the feature fusion process, we introduce an adaptive weighted fusion strategy. This strategy dynamically adjusts the weights of features from various perspectives, ensuring each feature maximizes its contribution, thereby enhancing classification performance.
3) We conduct comprehensive experiments on four widely used benchmark datasets to assess the effectiveness of the proposed model. The results illustrate that, in comparison to existing methods, our approach achieves superior performance in both classification accuracy and generalization capability.

The main differences between the proposed KDMSC method and other knowledge distillation-based methods are as follows.

1) Existing knowledge distillation-based methods [32], [33] primarily process HSI directly without transforming it into a graph structure. To further explore the intrinsic structure and information of HSI, the KDMSC method innovatively applies graph knowledge distillation technology to HSI classification for the first time, advancing research in this field.
2) In existing knowledge distillation-based methods [34], [35], the student model is typically of a single structure. In contrast, the KDMSC method significantly enhances the model's learning capacity and classification performance by introducing a dual-student model architecture,

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thereby improving the overall model's expressive power and generalization ability.

The remainder of this article is organized as follows. Section II provides an overview of related work on our research problem. Section III presents a detailed description of the proposed KDMSC method. Section IV presents a thorough analysis of the experimental results. Section V discusses the model. Finally, Section VI concludes this article.

II. Related Works

This section will focus on the historical development of deep learning-based methods for HSI classification in few-shot scenarios, providing a systematic review of key achievements in this field.

In HSI classification, data annotation typically requires domain experts to manually label samples based on specific application contexts. This process is not only time-consuming and labor-intensive but also demands substantial human and material resources. To address this challenge, researchers have actively explored strategies such as transfer learning and FSL to reduce the reliance of models on large amounts of labeled data while enhancing classification accuracy and generalization capability.

Transfer learning [36] refers to a method that facilitates specific tasks in the target domain by transferring models or knowledge acquired from the source domain. In this field, Yang et al. [37] were the first to apply transfer learning techniques to enhance HSI classification performance in few-shot scenarios, pioneering a new research direction. To address the challenge of limited labeled samples in HSI classification, Guan and Lam [38] proposed an innovative cross-domain contrastive learning (XDCL) framework. This framework extracts shared features between the spectral and spatial domains through cross-domain discrimination tasks while mitigating the interference of irrelevant information, thereby enhancing the semantic significance and generalization capability of representations. In [39], a novel deep cross-domain FSL (DCFSL) approach was introduced, which, for the first time, integrates FSL and domain adaptation into a unified framework to improve model generalization and interpretability. Furthermore, to tackle the domain adaptation challenge between the source and target domains, Zhang et al. [40] proposed a graph-information aggregation-based cross-domain FSL (Gia-CFSL) framework. By leveraging intradomain and cross-domain graph alignment strategies, this method effectively mitigates the adverse impact of domain shifts on classification performance. S4DL [41] disentangles spatial–spectral features based on domain-shift intensity to enhance cross-domain robustness. In addition, MLPA [42] performs hierarchical feature alignment within the feature extraction network to achieve fine-grained alignment between the source and target domains. By introducing prototype-based adversarial training, MLPA enhances discriminability and reduces negative transfer, significantly improving the generalization performance of cross-domain few-shot HSI classification. Recent breakthroughs in feature disentanglement and spectral coordinate modeling have emerged: Qin et al. [43] proposed an orthogonal low-rank feature disentanglement (FDFSL) method, which employs multi-order interactive central difference (MICD) blocks and a data-driven self-distillation scheme to significantly suppress source-domain bias while enhancing target-domain knowledge acquisition; Li et al. [44] pioneered the spectral coordinate Transformer (SCFormer), utilizing rotary position embedding (RoPE) to preserve spectral coordinate priors and designing an orthogonal complement space projection (OCSP) loss function to improve intradomain consistency. In addition, to extract comprehensive semantic information from multiple auxiliary tasks and efficiently integrate multitask learning into a unified network architecture, Liu et al. [45] introduced a hybrid multitask learning (HyMuT) framework with cross-task shared representations, further advancing the field of HSI classification.

FSL aims to achieve accurate classification of target data using a limited number of labeled samples, thereby effectively reducing dependence on large-scale annotated datasets. Currently, FSL methods can be broadly categorized into three main types: prototype networks, relation networks, and Siamese networks.

Prototype networks classify new samples by generating a prototype vector for each class in the embedding space and determining class membership based on the distance between new samples and these prototype vectors [46]. In [47], a global prototype network (GPN) was proposed, integrating global representation learning with few-shot classification. This model learns global prototype representations in the embedding space and employs a nearest-neighbor classifier to perform HSI classification under few-shot scenarios, significantly improving classification performance. To enhance spectral–spatial information learning, Xi et al. [48] introduced a deep prototype network that incorporates a hybrid residual attention module and a discriminative distance loss. This approach effectively captures spectral–spatial information, enhances feature compactness, and improves classification accuracy. In [49], a cross-scale graph prototype network (X-GPN) was proposed, integrating multiscale graph convolution and a self-branch attention-weighted module to enhance HSI classification performance. In [50], a novel active learning-based prototype network (ALPN) was introduced, which actively selects samples and optimizes feature extraction, significantly improving classification performance in few-shot scenarios. Furthermore, to enhance classification accuracy and sample selection precision, Ding et al. [51] proposed an active learning-based prototype learning with graph convolution network (PLGCN). This method iteratively optimizes class prototypes and leverages a graph convolutional network to extract the spatial distribution features of unlabeled samples, thereby improving classification performance. Xie et al. [52] introduced a feature consistency-based prototype network (FCPN), specifically designed for open-set HSI classification tasks, where the class distributions in the training and testing phases are inconsistent. This method optimizes feature consistency to mitigate the impact of distribution mismatches, further improving classification robustness. Finally, Li et al. [53] proposed semantically guided prototype learning (SGPL), which innovatively integrates class semantic information with

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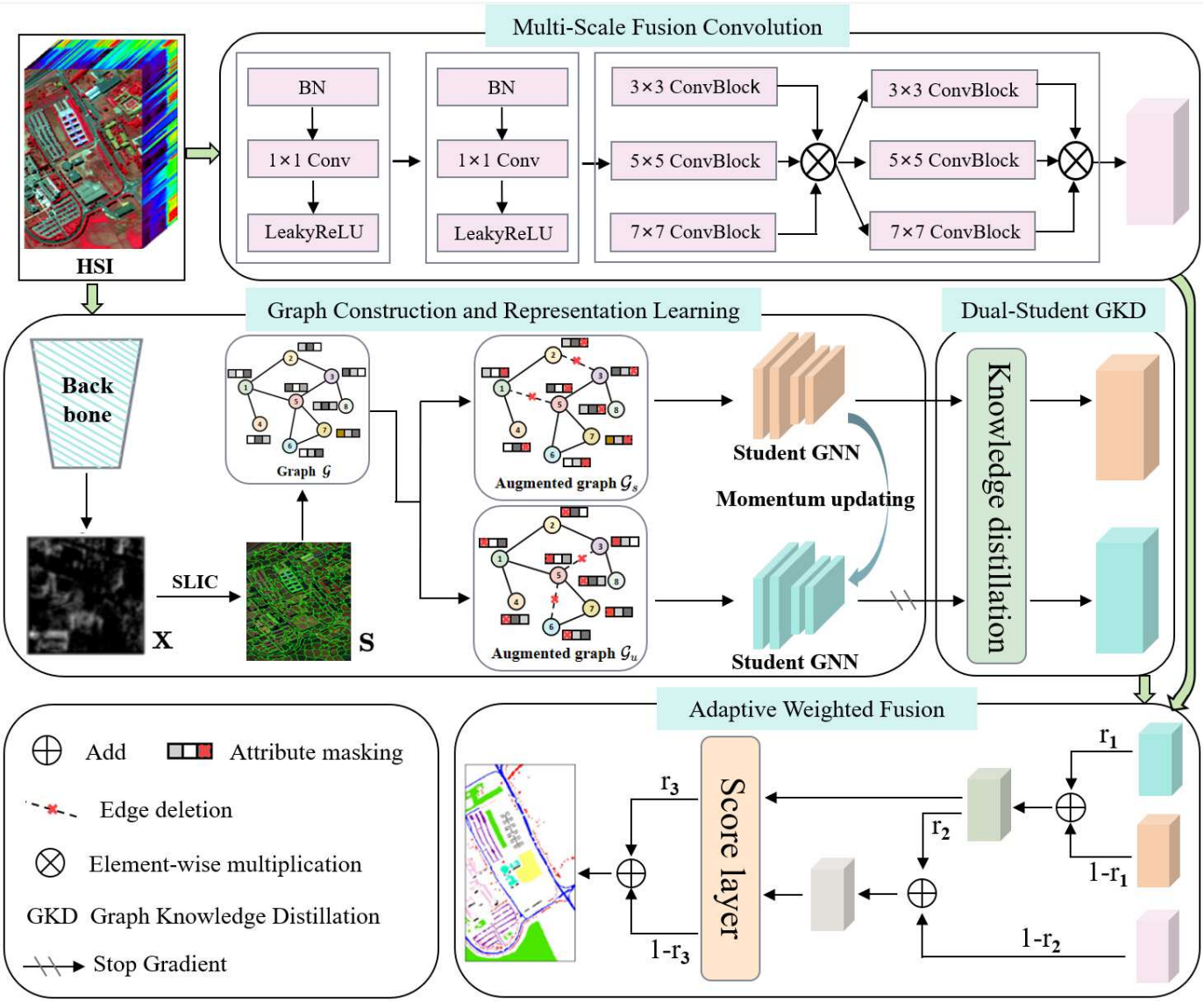


Fig. 1. Overall architecture of the proposed KDMSC method. The framework consists of four core modules: MSFC, graph construction and representation, DSGKD, and AWF. First, the original HSI is simultaneously fed into both the MSFC and graph construction and representation modules. The MSFC module performs denoising using 1-D convolution, followed by 2-D convolutions with different receptive fields to extract pixel-level features $\mathbf{H}_c$. The graph construction and representation module extracts feature map $\mathbf{X}$ via a backbone network and then applies the SLIC algorithm on $\mathbf{X}$ to generate a superpixel map $\mathbf{S}$. Based on $\mathbf{S}$, a spatial graph $\mathcal{G}$ is constructed, and enhanced graphs $\mathcal{G}_s$ and $\mathcal{G}_u$ are generated through graph augmentation. These graphs are input separately into student GNNs with identical architectures but independent parameters for feature extraction. The extracted features are processed by the DSGKD module to obtain two graph features $\mathbf{H}_s$ and $\mathbf{H}_u$. Finally, the pixel-level feature $\mathbf{H}_c$ from MSFC and the two graph features $\mathbf{H}_s$ and $\mathbf{H}_u$ from DSGKD are adaptively fused in the AWF module. This module learns the weight relationships among different features and automatically assigns appropriate fusion ratios. The fused comprehensive representation not only balances the spectral details and spatial structure information but also enhances the discriminative capability of the features, which is then fed into the classification head to generate a high-precision classification map.

spectral features to construct enhanced prototype generators. Through orthogonal feature space constraints, this approach significantly improves the robustness and generalization capability for cross-domain few-shot classification.

Relation networks classify samples by analyzing the similarity or correlation between the support set and the query set [54]. Gao et al. [55] designed a novel deep classification model based on a relation network, incorporating a meta-learning paradigm for training. This model aims to accurately classify new HSI using a limited number of labeled samples. To enhance model efficiency, Wang et al. [56] proposed an improved deep separable relation network, which symmetrically concatenates the feature vectors of support and query samples. This design effectively captures intersample similarity while reducing dependence on labeled samples, thereby improving training efficiency. In [57], a spectral-semantic transformation FSL (SSTFSL) method is introduced, which effectively extracts both semantic and spectral details for cross-domain few-shot HSI classification. To further enhance classification performance in few-shot HSI scenarios, Zeng et al. [58] proposed a multistage relation network with dual-metric (DM-MRN). This method employs a dual-metric mechanism and an adaptive weighting strategy to improve classification accuracy with limited labeled samples. Finally, to more comprehensively explore and utilize intersample relationships, Liu et al.

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[59] introduced a method that learns sample relations at different hierarchical levels and incorporates them into the model learning process, thereby optimizing cross-domain few-shot HSI classification.

Siamese networks are a type of neural network architecture composed of multiple identical subnetworks that share parameters and work collaboratively [60]. Siamese networks are commonly used to measure the similarity or dissimilarity between input samples [61] and have been widely applied in areas such as image retrieval [62], [63], speech recognition [64], [65], and object tracking [66], [67]. Huang and Chen [68] proposed a novel architecture named dual-path Siamese CNN (Dual-SCNN), which enhances HSI classification performance by combining expanded morphological features, a Siamese network, and spectral–spatial feature fusion. To improve the model's adaptability in unfamiliar scenarios, Wang et al. [69] introduced a meta-learning hyperspectral object detection method based on a Siamese network (MLSN). This approach effectively enhances the model's generalization performance by designing a three-channel and two-channel structure integrated with meta-learning strategies. In [70], an innovative method called 3DAES is proposed, which embeds the Siamese network within a semi-supervised learning framework to significantly improve HSI classification performance under limited labeled samples. Furthermore, to alleviate overfitting issues during training, Xue et al. [71] proposed a novel spectral–spatial Siamese network (S3Net). This network employs a dual-branch structure and weighted contrastive loss to effectively enhance the model's discriminative ability while preventing overfitting during training.

III. Proposed Method

This section offers a comprehensive overview of the proposed methodology. Initially, a brief description of the model's overall framework is provided (see Section III-A). Following this, the process of graph construction and representation learning strategy is outlined (see Section III-B), along with the dual-student graph knowledge distillation (DSGKD) module (see Section III-C), the multiscale fusion convolution (MSFC) module (see Section III-D), the adaptive weight fusion (AWF) module (see Section III-E), and the ensemble learning module (see Section III-F) in a sequential manner. Finally, the training procedure of the model is detailed (see Section III-G).

A. Overview

This section provides a detailed introduction to the proposed KDMSC framework. As illustrated in Fig. 1, the framework consists of four key modules: graph construction and representation learning, DSGKD, MSFC, and AWF. From the superpixel perspective, the graph construction and representation module initially extracts features from the original HSI via a backbone network, resulting in a feature map $\mathbf{X}$. The simple linear iterative clustering (SLIC) algorithm [72] is then applied to perform superpixel segmentation, producing a segmentation map $\mathbf{S}$, which is used to construct a superpixel graph $\mathcal{G}$. Graph augmentation is subsequently conducted on $\mathcal{G}$, creating two augmented graphs ($\mathcal{G}_s$ and $\mathcal{G}_u$). Two student GNNs, with identical architectures but different parameters, are employed for these two graph perspectives to extract features and obtain node representations. Within the DSGKD module, the dual-graph models are treated as two peer student models, and a gating mechanism is introduced to achieve interactive integration, ultimately forming a more expressive ensemble teacher model. The optimization of the teacher model enhances the student models, and conversely, the continuous improvement of the student models further refines the teacher model. From the pixel perspective, the MSFC module first denoises the original HSI, followed by cross-scale convolutions to enable more precise and comprehensive pixel-level feature extraction. Once features are obtained from both perspectives, the AWF module dynamically adjusts the weights of features from different perspectives to achieve effective multiview feature fusion. Finally, ensemble learning is utilized to mitigate the impact of hyperparameter settings on classification performance.

It is worth emphasizing that the proposed KDMSC framework explicitly considers both spectral and spatial information. From the spectral perspective, the MSFC module utilizes 1-D convolution for denoising and multiscale 2-D convolutions to comprehensively extract fine-grained pixel-level spectral features. From the spatial perspective, the graph construction module builds a superpixel-based graph and employs GNN with dual-student distillation to capture global spatial correlations across different regions. Finally, the AWF module serves as a bridge that adaptively integrates pixel-level spectral features with graph-based spatial features, thereby achieving a more balanced and complementary spectral–spatial representation. This design ensures that the model not only leverages the discriminative spectral signatures of hyperspectral pixels but also fully exploits the structural dependencies among spatial regions.

B. Graph Construction and Representation Learning Module

HSI contains high-dimensional spectral information and fine-grained spatial details. Directly modeling the raw data not only significantly increases computational complexity but also introduces substantial redundant information, potentially compromising model efficiency and performance. To address this issue, this module designs a backbone network composed of two 2×2 convolutional layers for feature extraction and preprocessing of HSI. By inputting the raw HSI $\mathbf{F} \in \mathbb{R}^{H\times W\times C}$ into this network (where H, W, and C represent the height, width, and number of channels, respectively), the backbone effectively extracts salient features that are highly relevant to subsequent tasks while suppressing noise and redundant information. The convolution operation is formulated as follows:

$$\mathbf{X}^{l}_{i,j,m} = \delta\left(\sum_{c=0}^{C-1}\sum_{h=0}^{K-1}\sum_{w=0}^{K-1}\mathbf{W}^{l}_{h,w,c,m}\cdot\mathbf{X}^{l-1}_{i+h,j+w,c} + \mathbf{B}^{l}_{m}\right) \tag{1}$$

where $\mathbf{X}^{l}_{i,j,m}$ represents the output feature at spatial location (i, j) and channel m in the lth layer, $\mathbf{W}^{l}_{h,w,c,m}$ denotes the weight of the mth convolutional kernel at channel c in the lth layer, with a kernel size of $K \times K$, and $\mathbf{B}^{l}_{m}$ is the bias term. $\mathbf{X}^{l-1}_{i+h,j+w,c}$ corresponds to the input feature at spatial location $(i+h, j+w)$

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and channel c in the $(l-1)$th layer, where h and w represent the sliding window offsets of the convolutional kernel over the input feature map, and $\mathbf{X}^0 = \mathbf{F}$. The final activation function $\delta(\cdot)$ is the ReLU function applied to the $K \times K$ convolution operation.

After preprocessing the HSI, the pixels continue to display complex spectral–spatial characteristics that are challenging to fully represent in the traditional Euclidean space. By employing graph structures for modeling, these non-Euclidean properties can be effectively captured, thereby uncovering the intricate relationships among the objects. To establish the graph structure for the HSI, this section utilizes the SLIC algorithm for superpixel segmentation. The SLIC algorithm groups neighboring pixels with similar features into compact superpixel regions, which significantly reduces the number of nodes in the graph and thereby lowers the computational complexity. This process also maintains the spectral consistency and spatial correlation within local regions, laying a robust foundation for subsequent classification tasks.

The processed feature map is denoted as $\mathbf{X}$. The superpixel segmentation method is applied to generate the superpixel segmentation map $\mathbf{S}$, and the superpixel descriptor matrix is $\mathbf{V} = (\mathbf{v}_1, \mathbf{v}_1, \ldots, \mathbf{v}_Z) \in \mathbb{R}^{Z \times C}$, where Z denotes the number of superpixels. Specifically, the affinity matrix $\mathbf{Q} \in \mathbb{R}^{HW \times Z}$ represents the mapping relationship between pixels and superpixels, where Z is the number of superpixels, and is computed as follows:

$$\mathbf{Q}_{ij} = \begin{cases} 1, & \text{if } \hat{\mathbf{X}}_i \in \mathbf{S}_j \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

where $\hat{\mathbf{X}} = \text{Flatten}(\mathbf{X})$, $\hat{\mathbf{X}}_i$ represents the ith pixel in $\mathbf{X}$, and $\mathbf{S}_j$ denotes the jth superpixel. Subsequently, feature transformation is performed using an encoder and decoder, which is computed as follows:

$$\mathbf{V} = \text{Encoder}(\mathbf{X}, \mathbf{Q}) = \hat{\mathbf{Q}}^{\mathrm{T}} \hat{\mathbf{X}} \tag{3}$$

$$\mathbf{X}' = \text{Decoder}(\mathbf{V}, \mathbf{Q}) = \text{Reshape}(\mathbf{Q}\mathbf{V}) \tag{4}$$

where $\hat{\mathbf{Q}}^{\mathrm{T}}$ represents the transpose of the matrix $\mathbf{Q}$ after normalizing its columns, $\mathbf{V}$ is the feature matrix of the superpixels, and $\mathbf{X}'$ is the pixel-level feature derived from the superpixel-level feature transformation.

After obtaining the superpixel descriptor matrix $\mathbf{V}$, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is constructed. Each node in the graph $\mathcal{G}$ corresponds to a different superpixel, and the edge relationships between nodes in $\mathcal{G}$ are constructed based on the spatial adjacency relationships between superpixels. Specifically, in the feature matrix $\mathbf{V}$ of graph $\mathcal{G}$, each element represents the feature of a superpixel; for the adjacency matrix $\mathbf{A}$, the element in the ith row and jth column is computed as follows:

$$\mathbf{A}_{ij} = \begin{cases} 1, & \text{if } \mathbf{v}_i \in \mathcal{N}(\mathbf{v}_j) \text{ or} \mathbf{v}_j \in \mathcal{N}(\mathbf{v}_i) \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where $\mathcal{N}(\mathbf{v}_j)$ and $\mathcal{N}(\mathbf{v}_i)$ represent the neighbor sets of superpixels $\mathbf{v}_j$ and $\mathbf{v}_i$, respectively.

In HSI classification tasks where labeled samples are limited, relying on a single graph structure is insufficient to fully capture the complex relational information inherent in the data. This limitation hampers the model's ability to deeply explore the features, thereby compromising its classification performance. To overcome this challenge, we propose a multiview graph representation strategy designed to compensate for the shortcomings of a single graph structure and to further uncover the latent characteristics of HSI data.

Specifically, by introducing random perturbations to the node features and edges of the graph $\mathcal{G}$ at a certain proportion, two augmented graph perspectives, $\mathcal{G}_s$ and $\mathcal{G}_u$, are generated. For the random perturbation of node features, we apply a masking ratio of ns% to randomly mask the attributes in the feature matrix $\mathbf{X}$ of graph $\mathcal{G}$, setting the masked attributes to zero, thus obtaining two new feature matrices, $\mathbf{X}_s$ and $\mathbf{X}_u$. For the random perturbation of edges, we apply an edge deletion ratio of es% to randomly remove edges in graph $\mathcal{G}$, resulting in two new adjacency matrices, $\mathbf{A}_s$ and $\mathbf{A}_u$.

Furthermore, $\mathcal{G}_s$ and $\mathcal{G}_u$ are fed into two distinct student GNN models. The message passing process of the lth layer in the GCN of the student model is formalized as follows:

$$\mathbf{H}^{l+1} = \text{BN}\left(\text{ReLU}\left(\hat{\mathbf{D}}^{-1/2}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-1/2}\right)\mathbf{H}^l\mathbf{W}^l\right) \tag{6}$$

where BN denotes the batch normalization layer, which accelerates training and helps mitigate overfitting. ReLU is the activation function, and $\mathbf{W}^l$ is the trainable parameter matrix at the lth layer. Note that $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, where $\mathbf{A}$ is the adjacency matrix of the current graph, $\mathbf{I}$ is the identity matrix and $\hat{\mathbf{D}}_{ii} = \sum_{j=1}^{Z} \hat{\mathbf{A}}_{ij}$. The features at the $(l+1)$th layer for the two student models can then be represented as $\mathbf{H}_s^{l+1}$ and $\mathbf{H}_u^{l+1}$, respectively.

To enhance the model's generalization capability, we integrate GNN with the mean teacher algorithm [73] and design two student networks with identical architectures but different parameters. Specifically, in each training iteration, one student network optimizes its parameters through gradient descent, while the other updates its parameters using a momentum-based mechanism and exponential moving average (EMA). This design not only significantly reduces instability during the model's training process but also improves the model's performance in few-shot scenarios, thereby more efficiently leveraging the complementary properties between multiview graphs. This, in turn, provides more reliable feature representations for subsequent DSGKD. The formulation is as follows:

$$\theta_k^u = p\theta_{k-1}^u + (1-p)\,\theta_k^s \tag{7}$$

where θ_k^u represents the weight parameters of the second student model at the kth training iteration, θ_k^s denotes the weight parameters of the first student model at the kth training iteration, and p is the momentum coefficient.

C. Dual-Student Graph Knowledge Distillation Module

In FSL scenarios, the lack of clear signals to guide the learning process often results in poor generalization and robustness for models that directly learn from unlabeled data. Graph knowledge distillation provides an effective remedy by transferring knowledge from a teacher model to a student model, thereby enabling the student model to extract valuable information from the unlabeled data. In this section, we leverage DSGKD to enhance the performance of HSI classification

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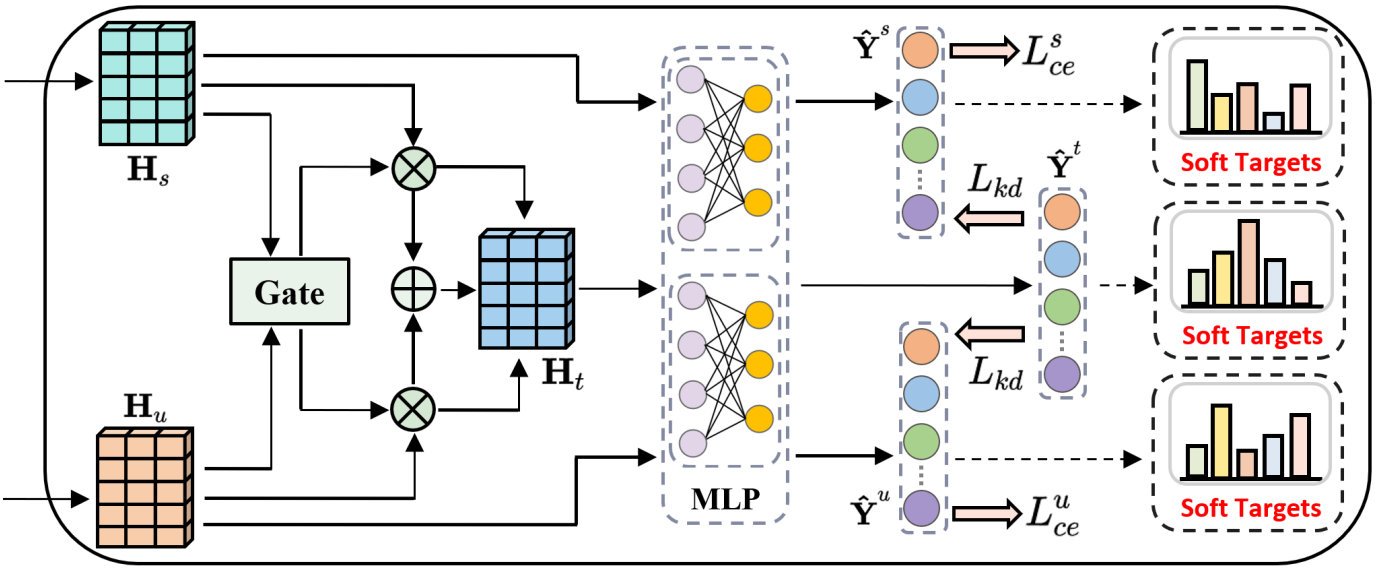


Fig. 2. DSGKD architecture.

in few-shot settings. Specifically, we treat two graph models as peer student models and integrate them to create a more robust teacher model. The teacher model's predictions for all superpixels are then used as additional supervisory signals to refine the student models' modeling capabilities. This iterative process of improvement between the student and teacher models leads to a more effective integrated teacher model. The entire process is depicted in Fig. 2.

Following the steps in Section III-B, we independently model the superpixel node embeddings in the two student models, resulting in two distinct superpixel feature representation matrices, $\mathbf{H}_s$ and $\mathbf{H}_u$. To effectively fuse the diverse representations from both student graphs, we design a gating mechanism that leverages the integrated semantic cues to adaptively weight the contributions of $\mathbf{H}_s$ and $\mathbf{H}_u$. As the superpixel embeddings implicitly encode class-specific information, the learned gating coefficients can capture the semantic and categorical differences between the two representations, thereby enhancing the fusion process and facilitating the construction of a more expressive teacher model. The formal definition is as follows:

$$\mathbf{H}_t = \mathbf{g} \odot \mathbf{H}_s + (1 - \mathbf{g}) \odot \mathbf{H}_u \tag{8}$$

where the gating coefficient $\mathbf{g}$ is given by the following formula:

$$\mathbf{g} = \delta\left(\mathbf{W}_g \left[\mathbf{H}_s, \mathbf{H}_u\right] + \mathbf{b}_g\right) \tag{9}$$

where $\mathbf{W}_g$ and $\mathbf{b}_g$ are the learnable weight matrix and bias vector, and δ is the activation function. It is worth noting that, although the gating mechanism is not explicitly conditioned on class information during modeling, its input feature representations $\mathbf{H}_s$ and $\mathbf{H}_u$ inherently encode discriminative interclass differences. Consequently, the gating mechanism can automatically capture class-related representation discrepancies by modeling the concatenated features, enabling it to adaptively assign more appropriate weights to the more reliable student model in different spatial regions. This is particularly effective in areas with significant class distinctions or ambiguous class boundaries.

After obtaining the superpixel representations $\mathbf{H}_t$ of the teacher model through the gate mechanism, we further process $\mathbf{H}_s$, $\mathbf{H}_u$, and $\mathbf{H}_t$ using (10) to derive the corresponding superpixel logits, denoted as $\mathbf{Z}_s$, $\mathbf{Z}_u$, and $\mathbf{Z}_t$, respectively,

$$\mathbf{Z} = \mathbf{W}_0\mathbf{H} + \mathbf{b}_0 \tag{10}$$

where $\mathbf{W}_0$ and $\mathbf{b}_0$ are the learnable weight matrices and bias vectors.

To distill the knowledge of the teacher model back into the student model, we first compute the soft label probabilities of both the student and teacher models with a temperature hyperparameter γ

$$\begin{aligned}\hat{\mathbf{Y}}^s &= \frac{1}{1 + e^{-\mathbf{Z}_s/\gamma}} \\ \hat{\mathbf{Y}}^u &= \frac{1}{1 + e^{-\mathbf{Z}_u/\gamma}} \\ \hat{\mathbf{Y}}^t &= \frac{1}{1 + e^{-\mathbf{Z}_t/\gamma}}.\end{aligned} \tag{11}$$

To account for the label discrepancies between the soft labels of the two student models and the labeled superpixels, we employ the cross-entropy loss function to compute two separate losses, L^s_{ce} and L^u_{ce}, as formulated as follows:

$$L^s_{\text{ce}} = -\frac{1}{Z}\sum_{i=1}^{Z}\sum_{j=1}^{L}\mathbf{Y}^s_{ij}\log\left(\hat{\mathbf{Y}}^s_{ij}\right) \tag{12}$$

$$L^u_{\text{ce}} = -\frac{1}{Z}\sum_{i=1}^{Z}\sum_{j=1}^{L}\mathbf{Y}^s_{ij}\log\left(\hat{\mathbf{Y}}^u_{ij}\right) \tag{13}$$

where Z, L, and $\mathbf{Y}^s$ denote the number of superpixels, the number of categories, and the superpixel-level label map, respectively.

To fully leverage the information from unlabeled samples, we design a knowledge distillation loss (L_{kd}), which aims to

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guide the learning process of the student model through the knowledge of the teacher model. The loss is computed as follows:

$$L_{\mathrm{kd}} = \frac{1}{Z}\sum_{i=1}^{Z} \|\mathbf{Y}_i^t - \mathbf{Y}_i^s\|_1 - \|\mathbf{Y}_i^t - \mathbf{Y}_i^u\|_1. \tag{14}$$

Finally, the total loss of DSGKD is formulated as follows:

$$L_{\mathrm{ds}} = L_{\mathrm{ce}}^s + L_{\mathrm{ce}}^u + \lambda L_{\mathrm{kd}} \tag{15}$$

where λ is the corresponding weighting factor.

D. MSFC Module

The preceding discussion has mainly centered on the superpixel perspective for extracting spectral–spatial features of HSI, with an emphasis on global information modeling. However, due to the inherent scale limitations of superpixel segmentation, comprehensively capturing finer grained local features presents a significant challenge. Although multiple scales of superpixel segmentation are subsequently employed and integrated through ensemble learning, deficiencies in fine-grained feature representation persist. To address the lack of local pixel-level information, this section proposes a multi-scale fusion convolutional approach. By refining the extraction of local features, this method enhances the representation of fine-grained details, thereby facilitating a collaborative modeling framework that integrates both global and local spectral–spatial information. This integrated approach ultimately leads to a more comprehensive improvement in classification performance.

Although raw HSI data contain rich spectral information, they are inevitably accompanied by significant noise. Such noise not only interferes with the extraction of effective features but may also introduce biases in classification results, ultimately reducing the classification accuracy of the model. To address this issue, we first introduce a denoising module composed of two 1×1 convolutional layers to perform noise reduction. Specifically, by inputting the raw HSI $\mathbf{F}$ into a spectral convolutional layer, spectral features are progressively extracted and refined, leading to the following representation:

$$\mathbf{X}_{i,j,m}^l = \delta\left(\sum_{c=0}^{C-1} \mathbf{W}_{i,j,m,c}^l \mathrm{BN}\left(\mathbf{X}_{i,j,c}^{l-1}\right) + \mathbf{B}_{i,j,m,c}^l\right) \tag{16}$$

where $\mathbf{X}_{i,j,m}^l$ denotes the output feature at spatial position (i, j) and channel index m in the lth layer, while $\mathbf{X}_{i,j,c}^{l-1}$ represents the input feature at spatial position (i, j) and channel index c in the $(l-1)$th layer, and $\mathbf{X}^0 = \mathbf{F}$. $\mathbf{W}_{i,j,m,c}^l$ and $\mathbf{B}_{i,j,m,c}^l$ correspond to the weight and bias of the 1×1 convolutional layer, respectively. BN denotes the batch normalization layer, and $\delta(\cdot)$ represents the ReLU activation function.

To effectively extract local pixel-level features, we design a cross-multiscale convolution module. This module consists of two multiscale convolutional layers, each employing convolutional kernels of three different sizes: 3×3, 5×5, and 7×7. Within each multiscale convolutional layer, the input features are first mapped through convolution operations with different kernel sizes. The extracted features from different kernels are then aggregated to achieve effective cross-scale feature fusion. The formal representation is given in (1), where K takes values of 3, 5, and 7, respectively. Ultimately, the pixel-level features $\mathbf{H}_c$ are obtained through MSFC.

Through the aforementioned design, the MSFC module can capture feature information across different receptive fields, effectively extracting fine-grained pixel-level details and contextual correlations. This enables the provision of richer and more accurate feature representations for subsequent tasks.

E. AWF Module

To effectively integrate multiview features and enhance the expressive capability of the model, this section introduces the AWF method. AWF consists of two modules: the feature-weighted fusion module and the score-weighted fusion module.

1) Feature-Weighted Fusion: To further enhance the diversity and robustness of feature representation, we first perform adaptive fusion of the global-level superpixel features $\mathbf{H}_s$ and $\mathbf{H}_u$. Specifically, a learnable weight parameter γ_1 is introduced to dynamically adjust the contribution of each feature during the fusion process, achieving a weighted summation of the two feature streams. This adaptive fusion strategy effectively integrates the advantages of both features, further improving the precision and robustness of global-level feature representation. The mathematical formulation of the fusion process is defined as follows:

$$\mathbf{H}_{su} = \gamma_1 \cdot \mathbf{H}_s + (1 - \gamma_1) \cdot \mathbf{H}_u. \tag{17}$$

Next, we further fuse the globally fused features $\mathbf{H}_{su}$ with the locally pixel-level features $\mathbf{H}_c$. Specifically, by defining a learnable weight parameter γ_2, these two features are adaptively combined through weighted summation. This design enables the global features to provide macro-level information while retaining the detailed expression capabilities of local pixel features, thereby enhancing the model's expressive power and classification performance. The mathematical formulation of the fusion process is as follows:

$$\mathbf{H}_{suc} = \gamma_2 \cdot \mathbf{H}_{su} + (1 - \gamma_2) \cdot \mathbf{H}_c. \tag{18}$$

2) Score-Weighted Fusion: The obtained $\mathbf{H}_{su}$ and $\mathbf{H}_{suc}$ are, respectively, input into the scoring layer, where the features are transformed to obtain the soft label probabilities $\mathbf{P}_{su}$ and $\mathbf{P}_{suc}$ corresponding to $\mathbf{H}_{su}$ and $\mathbf{H}_{suc}$. The formal expression is as follows:

$$\begin{aligned} \mathbf{P}_{su} &= \mathrm{softmax}\left(\mathbf{W}_1\mathbf{H}_{su} + \mathbf{b}_1\right) \\ \mathbf{P}_{suc} &= \mathrm{softmax}\left(\mathbf{W}_2\mathbf{H}_{suc} + \mathbf{b}_2\right) \end{aligned} \tag{19}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are the weight matrices of the scoring layer, $\mathbf{b}_1$ and $\mathbf{b}_2$ are the corresponding bias vectors, and softmax denotes the normalization operation.

Finally, we perform adaptive weighting of the two probability distributions $\mathbf{P}_{su}$ and $\mathbf{P}_{suc}$, dynamically adjusting the weight of each distribution through a learnable parameter γ_3. This fully leverages the complementarity of the multiview features, as formalized in (20). Based on this, we use the cross-entropy loss function to optimize the fused probability

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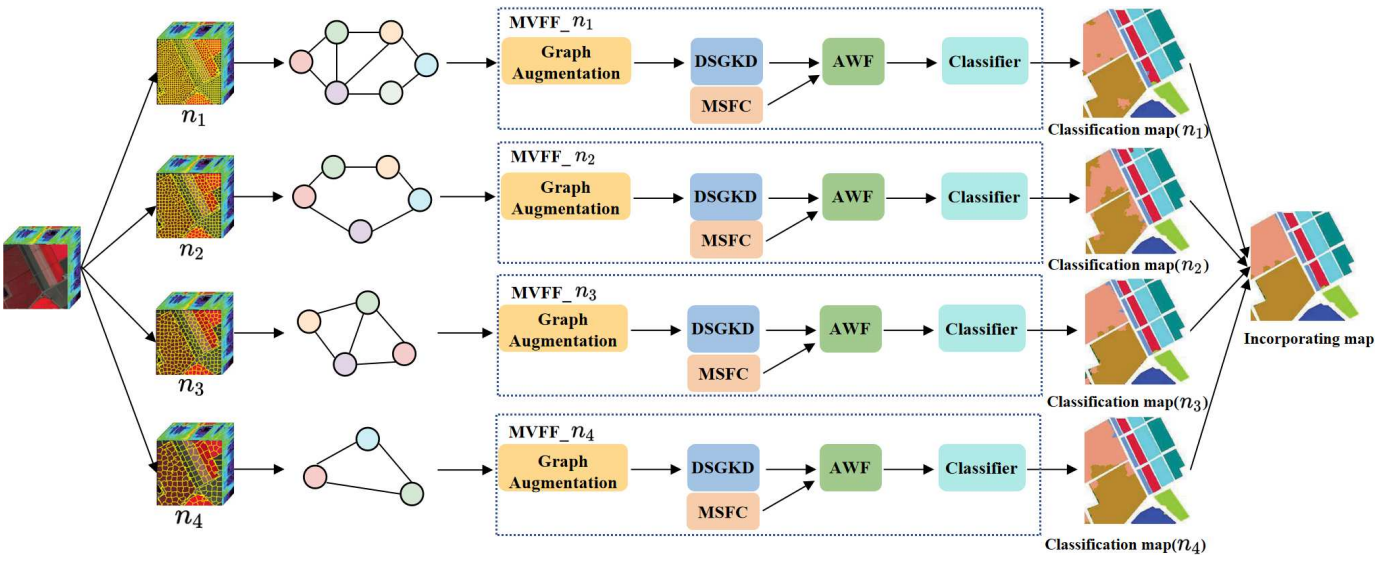


Fig. 3. Ensemble learning architecture.

distribution against the true labels. This effectively measures the discrepancy between the predicted and true distributions, and by optimizing the loss L_{awf}, it guides the model to progressively improve classification performance and converge to a more optimal solution. The formula is as shown in (21)

$$\mathbf{P} = \gamma_3 \cdot \mathbf{P}_{suc} + (1 - \gamma_3) \cdot \mathbf{P}_{su} \tag{20}$$

$$L_{\text{awf}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{L} \mathbf{Y}_{ij}^{c} \log \left(\mathbf{P}_{ij}\right) \tag{21}$$

where N denotes the total number of pixels, and $\mathbf{Y}^c$ represents the ground-truth label map of the pixels.

Through the AWF strategy, multiview features are integrated, effectively leveraging the complementarity of different perspectives, resulting in a more comprehensive fused representation. This dynamic weight adjustment mechanism not only adapts to the varying importance of different features in the task but also effectively mitigates issues of insufficient information or bias that may arise from relying on a single feature. As a result, it enhances the model's adaptability to complex data and improves classification accuracy.

F. Ensemble Learning

In the graph construction and representation module, the superpixel graph consists of Z superpixels, where the number Z is a key hyperparameter affecting segmentation accuracy and classification performance. Specifically, a smaller Z value may introduce pixels from different classes within the same superpixel, while a larger Z value increases the model's computational complexity and memory requirements. To better adapt to the multiscale characteristics of different land cover features in various HSIs, this article introduces n (representing the number of pixels in each superpixel) to indirectly determine the value of Z

$$Z = \frac{H \times W}{n}. \tag{22}$$

Due to the varying characteristics of each HSI dataset, the parameter n will be selected from four different values (n_1, n_2, n_3, n_4) based on its features. A detailed analysis will be provided in Section IV-C1. These four scales are applied to the HSI segmentation task, and the segmentation results at each scale are integrated using the majority voting method to determine the final segmentation outcome. Fig. 3 illustrates the ensemble learning framework.

G. Model Training

In the training phase of the model, we comprehensively consider the loss functions from different components and aggregate them to formulate the final optimization objective. The overall loss function can be expressed as follows:

$$L = L_{\text{awf}} + L_{\text{ds}} = L_{\text{awf}} + L_{\text{ce}}^{s} + L_{\text{ce}}^{u} + \lambda L_{\text{kd}} \tag{23}$$

where L_{awf} is the loss of the AWF module and L_{ds} is the loss of DSGKD. By optimizing the above total loss function, the model can effectively integrate the information from each loss term, improving generalization ability in the FSL scenario while maintaining high classification accuracy.

IV. Experiments and Results

In this section, we first describe the dataset utilized and the specific implementation details. Subsequently, a thorough performance evaluation of KDMSC is conducted, encompassing hyperparameter analysis, module ablation study, and generalization analysis. Following this, a comparative performance analysis between KDMSC and existing state-of-the-art methods is presented.

A. Datasets

Four benchmark HSI datasets, including Indian Pines, University of Pavia, Salinas, and Houston, are used in this study. The detailed descriptions, including data characteristics, sample splits, and corresponding figures and tables, are provided in Section I of the Supplementary Material.

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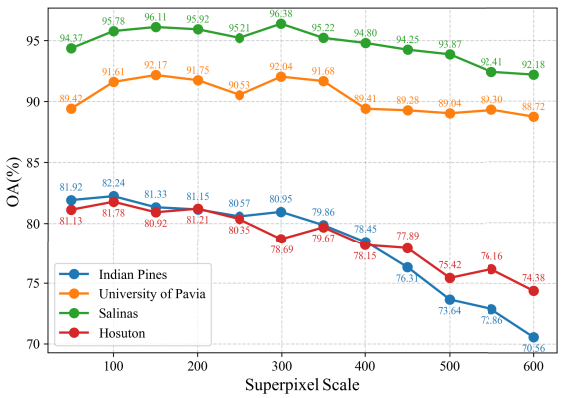


Fig. 4. Variation of model performance under different superpixel segmentation scales.

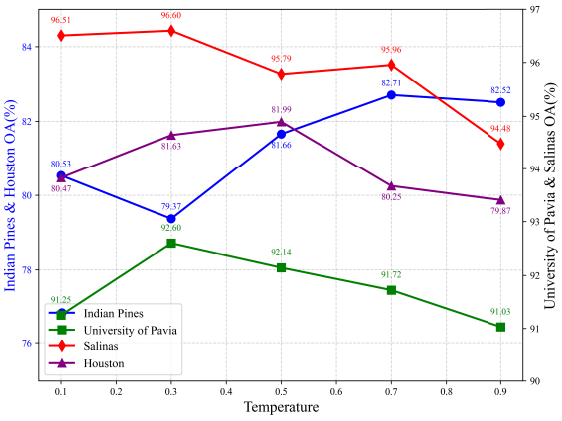


Fig. 5. Variation of model performance with different temperature parameters.

B. Experimental Settings

The proposed method, KDMSC, has been implemented using PyTorch 1.12.1, with model parameters optimized via the Adam optimizer. The experiments have been conducted on an NVIDIA GeForce RTX 3080 GPU under a Python 3.9 environment. In our experiments, to avoid the issue of information leakage during supervised learning, we follow the methods in previous research [15], [21] to generate the training and testing samples and ensure that they are strictly separated. Specifically, for each dataset, five labeled samples are randomly selected from each class to form the training set, while the remaining samples are allocated for testing. Pixels selected for training are completely excluded from the testing set, ensuring that there is no spatial overlap or duplicate pixels between the two sets. To ensure a fair evaluation, we perform ten random samplings for each dataset and conduct ten independent experiments, training the model for 200 iterations in each run. Performance is assessed using five metrics: overall accuracy (OA), average classification accuracy (AA), Kappa coefficient (Kappa), runtime, and the number of network parameters. In addition, the hyperparameters, learning rate r and λ, are set to 0.001 and 0.1, respectively.

To validate the effectiveness and generalizability of the KDMSC method, the following methods are selected for comparison: two representative CNN models (3DCNN [10] and SSRN [9]), two attention model (SPFormer [15] and HSI-MFormer [16]), three transfer learning methods (DCFSL [39], MLPA [42], and HyMuT [45]), one prototype network model (DPN-HRA [48]), one relation network model (RN-FSC [55]), and one Siamese network model (S3Net [71]). Unlike some existing methods that involve preprocessing steps such as band selection, outlier removal, or normalization, our method directly utilizes the original hyperspectral data without additional preprocessing. This ensures fair comparison under uniform input conditions. The compared methods all utilize the parameter settings specified in their original papers.

C. Hyperparameter Analysis

This section focuses on analyzing the impact of various hyperparameters on model performance from three key aspects. First, we examine the selection of the optimal superpixel segmentation scale for different datasets within the context of ensemble learning. Second, we discuss the regulatory influence of the temperature parameter on model performance. Finally, we investigate the effect of the enhancement ratio during the graph augmentation process on model performance. Figs. 4–6 illustrate the performance variations of the model across different parameter values.

1) Analysis of Superpixel Segmentation Scale: The size of the superpixel segmentation scale n has a significant impact on classification performance, as a fixed value is difficult to adapt to the diversity of different HSIs. In the ensemble framework, we conduct a statistical analysis of the range of n (from 50 to 600, with 12 scales in total) to determine the optimal segmentation scale for different datasets. Finally, as shown in Fig. 3, during the ensemble learning process, the classification results for the four optimal segmentation scales are integrated using a majority voting strategy.

As shown in Fig. 4, the optimal superpixel segmentation scales n differ across the datasets. Specifically, the optimal segmentation scales for the Indian Pines dataset are (50, 100, 150, and 200); for the University of Pavia dataset, they are (100, 150, 300, and 350); for the Salinas dataset, they are (100, 150, 200, and 300); and for the Houston dataset, the optimal scales are (50, 100, 150, and 200). This choice of segmentation scales is determined by the characteristics of each dataset, particularly in terms of spatial resolution and the complexity of land cover distribution. This variation is primarily due to differences in spatial resolution and the complexity of the land cover structures across different HSIs, which lead to varying sensitivities to segmentation scales. In practical applications, selecting an appropriate superpixel scale is closely related to the dataset's spatial resolution and land cover characteristics. Specifically, for datasets with low spatial resolution and complex land cover categories (e.g., Indian Pines and Houston), smaller segmentation scales (e.g., 50 and 100) are preferable for capturing fine-grained image details, as these datasets contain interwoven land cover types that require fine segmentation to distinguish boundaries. For high-resolution HSI datasets with highly heterogeneous land cover structures (e.g., University of Pavia), moderate segmentation scales (e.g., 150 and 300) strike a balance between noise suppression

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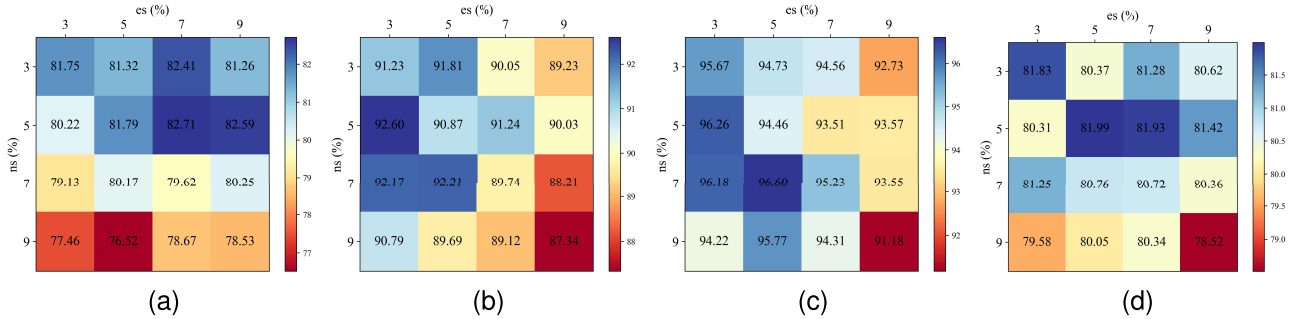


Fig. 6. Variation in model performance under different graph augmentation ratios. (a) Indian Pines. (b) University of Pavia. (c) Salinas. (d) Houston.

and detail preservation. In contrast, for datasets with high spatial resolution and more regular land cover shapes (e.g., Salinas), larger segmentation scales (e.g., 200 and 300) are beneficial in reducing redundant information and improving classification efficiency. Therefore, the selection of superpixel scales for different datasets is guided by a tradeoff among spatial resolution, land cover distribution density, and structural complexity. Ensemble learning, by integrating multiscale segmentation information, not only reduces the dependence on a single hyperparameter setting but also significantly enhances classification performance.

2) Analysis of the Temperature Parameter: In the knowledge distillation framework, different temperature parameters γ play a crucial role in the efficiency of knowledge transfer. This parameter controls the smoothness of the probability distribution of the soft labels output by the teacher model, directly affecting the learning process of the student model. To investigate the impact of the temperature parameter on model performance, this section presents a detailed analysis of model performance under different temperature settings, aiming to identify the optimal temperature hyperparameter configuration.

As shown in Fig. 5, different datasets exhibit varying responses to the temperature parameter γ during the knowledge distillation process. Specifically, the Indian Pines dataset achieves the highest classification accuracy of 82.71% when $\gamma = 0.7$; the University of Pavia and Salinas datasets achieve their best performances of 92.60% and 96.60% at $\gamma = 0.3$, respectively, while the Houston dataset reaches its optimal accuracy of 81.99% when $\gamma = 0.5$. This disparity primarily stems from the inherent characteristics of each dataset, particularly the differences in class similarity.

In the Indian Pines dataset, where the similarity between classes is relatively high, a higher temperature parameter effectively smooths the probability distribution of the soft labels, enhancing the information transfer and discriminability between classes. For the Houston dataset, where the class similarity is at a moderate level, a medium temperature parameter ($\gamma = 0.5$) facilitates a more balanced distillation effect by retaining class discriminability while enabling effective transfer of information from the soft labels. In contrast, in the University of Pavia and Salinas datasets, where there are fewer classes or more distinct class differences, a lower temperature parameter helps the student model directly learn more discriminative feature representations from the teacher model's deterministic outputs.

Therefore, the similarity between classes is a key factor in selecting the temperature parameter. From both theoretical and empirical perspectives, the optimal temperature value is positively correlated with class similarity: datasets with higher interclass similarity benefit from larger temperature values, while those with lower similarity are better suited to smaller temperature values. By adjusting the temperature parameter, optimal knowledge distillation performance can be achieved based on the varying class similarities. Experimental results demonstrate that selecting an appropriate temperature configuration, considering both dataset characteristics and class similarity, is crucial for improving the effectiveness of knowledge distillation.

3) Analysis of the Graph Augmentation Ratio: In this section, we conduct a statistical analysis to determine the optimal edge and node augmentation ratios for different datasets. Fig. 6 visually represents the performance variations across these datasets under various augmentation configurations. Specifically, the results for each dataset are as follows. Indian Pines achieves peak accuracy (82.71%) with 7% edge and 5% node augmentation. This dataset, characterized by spectrally similar agricultural classes, blurred boundaries, and small interclass differences, benefits from moderate edge augmentation. This helps to mitigate interference from overlapping classes and enhances the model's discriminative capability in boundary regions. University of Pavia reaches optimal accuracy (92.60%) with 3% edge and 5% node augmentation. Given its urban structures featuring distinct boundaries and compact layouts, minimal edge augmentation is most effective. This approach preserves the original graph relations and local spatial–textural integrity without introducing unnecessary perturbations. Salinas attains maximum accuracy (96.60%) with 5% edge and 7% node augmentation. The dataset's high spectral resolution and diverse, spectrally distinct categories warrant moderate augmentation. This strategy improves graph generalization and strengthens internode connections, thereby facilitating a better capture of interclass structural information. Houston exhibits the best performance (81.99%) with a balanced 5% edge and 5% node augmentation. Covering complex urban/nonurban mixtures with varied land cover and hybrid regions, this balanced approach reinforces structural representations in weakly connected areas. It enables effective

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TABLE I
ABLATION ANALYSIS OF DSGKD IN FOUR DATASETS

Gate	L_{ce}^{s}	L_{ce}^{u}	L_{kd}	Indian Pines	Pavia	Salinas	Houston
x	✓	✓	✓	80.42 ± 2.39	91.03 ± 2.71	94.81 ± 1.41	80.14 ± 1.07
✓	×	×	×	81.38 ± 1.27	90.15 ± 1.55	92.11 ± 0.89	79.46 ± 0.33
✓	×	✓	✓	81.55 ± 1.43	91.76 ± 2.31	95.12 ± 1.19	79.69 ± 2.24
✓	✓	×	✓	81.33 ± 2.46	91.89 ± 1.32	95.27 ± 2.01	80.23 ± 1.21
✓	✓	✓	×	80.62 ± 1.22	89.78 ± 1.29	94.35 ± 1.18	79.82 ± 1.92
×	×	×	×	79.03 ± 2.16	89.61 ± 1.34	92.77 ± 1.63	78.22 ± 1.54
✓	✓	✓	✓	82.71 ± 1.02	92.60 ± 0.99	96.60 ± 0.39	81.99 ± 1.75

modeling of cross-boundary relationships while preserving fine-grained spatial semantics.

The experimental results demonstrate that selecting the optimal augmentation ratio configuration based on the characteristics of different datasets is a crucial factor in improving model performance.

D. Ablation Study

To validate the effectiveness of the components in the proposed method, we have conducted a detailed analysis of DSGKD, MSFC, and AWF. The main findings are summarized as follows.

1) Effectiveness of DSGKD: To validate the effectiveness of DSGKD, this section evaluates the performance of key components in DSGKD, including the gate and the involved loss functions, across four benchmark datasets. The corresponding experimental results are detailed in Table I.

It is clear that removing the gate and simply summing the two feature streams leads to a decline in model performance. This decline can be attributed to the gate's ability to adaptively fuse the two feature streams, thereby more effectively extracting and utilizing key information. Further analysis reveals that removing either supervised loss, L_{ce}^{s} or L_{ce}^{u}, weakens the model's ability to leverage multiperspective features due to the complementary nature of the information extracted from labeled samples in the dual perspectives. In contrast, the performance degradation resulting from the removal of L_{kd} is more pronounced, particularly on the University of Pavia dataset, where the accuracy drops by 2.82%. This is primarily due to the insufficient utilization of unlabeled samples, which affects the effectiveness of knowledge distillation. Finally, the experimental results show that completely removing DSGKD significantly deteriorates model performance. Specifically, the accuracy decreases by 3.68% on the Indian Pines dataset, 2.99% on the Pavia dataset, and 3.83% on the Salinas dataset, and by 2.17% on the Houston dataset. These results strongly validate the effectiveness of DSGKD in enhancing model performance.

2) Effectiveness of MSFC: To validate the performance improvement of KDMSC with MSFC, this section compares five variants of KDMSC: KDMSC without the 1×1 convolution, KDMSC using only 3×3 single-scale convolutions, KDMSC using only 5×5 single-scale convolutions, KDMSC using only 7×7 single-scale convolutions, and KDMSC with MSFC. As shown in Fig. 7, KDMSC with MSFC outperforms all other variants across all datasets. Specifically, models without MSFC exhibit a decrease in classification accuracy by 0.98%–1.58% on the Indian Pines dataset, 0.21%–0.87% on the University of Pavia dataset, 0.32%–1.23% on the Salinas dataset, and 0.21%–1.08% on the Houston dataset. These experimental results demonstrate the effectiveness of MSFC.

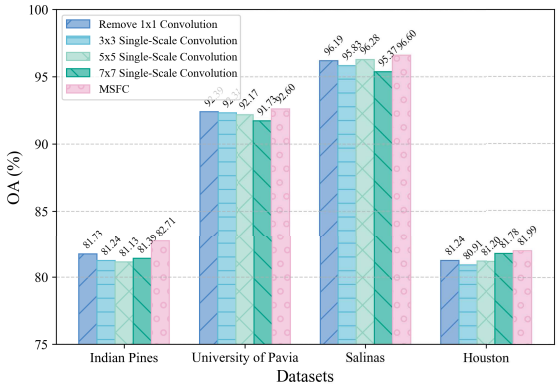


Fig. 7. Ablation analysis of MSFC in four datasets.

TABLE II
ABLATION ANALYSIS OF AWF IN FOUR DATASETS

	Indian Pines	Pavia	Salinas	Houston
Pixel View	76.14 ± 2.51	76.23 ± 1.87	88.79 ± 3.21	70.97 ± 1.24
S_pixel View	80.35 ± 1.86	87.51 ± 2.04	94.69 ± 2.42	78.04 ± 2.66
Honcat	80.29 ± 2.73	89.60 ± 2.11	95.16 ± 1.85	79.56 ± 2.08
Honcat1	81.94 ± 1.52	90.32 ± 1.55	94.93 ± 1.04	80.26 ± 1.75
Sscore	82.71 ± 1.02	92.60 ± 0.99	96.60 ± 0.39	81.99 ± 1.75

3) Effectiveness of AWF: To validate the effectiveness of AWF, this section conducts a comparative analysis of five variants of KDMSC. Specifically, "Pixel View" performs classification using pixel-level features extracted by the MSFC module, and "Superpixel View" performs classification using superpixel-level features generated by the DSGKD module. "Hconcat" denotes the direct concatenation of pixel-level and superpixel-level features for classification purposes. "Hconcat1" involves the weighted fusion of pixel-level and superpixel-level features using AWF for classification. "Sscore" further refines "Hconcat1" by incorporating score-based weighted fusion within AWF for more precise fusion and classification. The detailed experimental results are presented in Table II.

Clearly, superpixel-level features exhibit superior performance compared with pixel-level features in classification tasks, primarily due to the rich spatial–spectral features that can be extracted in the superpixel view by DSGKD. Nevertheless, the results obtained from "Hconcat" indicate that pixel-level features compensate for the loss of local pixel-level information, underscoring the complementary relationship between these two types of features. Furthermore, the incorporation of AWF in "Hconcat1" leads to an enhancement in classification performance, demonstrating the efficacy of feature-weighted fusion. Ultimately, "Sscore" achieves the highest classification accuracy through the utilization of score-based weighted fusion, thus establishing the superiority of this particular approach. In summary, the adaptive fusion of superpixel-level and pixel-level features effectively enhances the performance of HSI classification.

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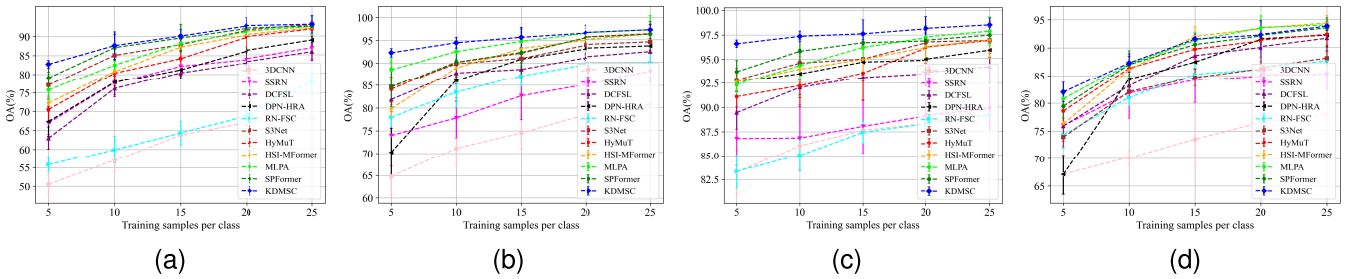


Fig. 8. Evolution of OA as a function of the number of labeled samples per class. (a) Indian Pines. (b) University of Pavia. (c) Salinas. (d) Houston.

TABLE III
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR INDIAN PINES

Class	3DCNN [10]	SSRN [9]	DCFSL [39]	DPN-HRA [48]	RN-FSC [55]	S3Net [71]	HyMuT [45]	HSI-MFormer [16]	MLPA [42]	SPFormer [15]	KDMSC
1	80.00±18.28	99.02±1.62	97.32±3.53	45.39±23.72	88.78±11.08	**100.00±0.00**	98.05±2.84	**100.00±0.00**	**100.00±0.00**	99.51±0.98	98.91±1.09
2	31.46±8.88	46.23±16.52	40.44±7.11	**75.96±14.55**	33.65±9.68	58.84±13.30	60.51±9.87	34.35±3.08	62.20±10.46	61.52±13.07	74.61±6.34
3	34.13±8.96	63.60±8.80	45.62±11.09	57.89±9.64	27.89±8.09	63.68±6.74	65.50±10.42	71.20±5.24	57.07±10.61	**71.25±9.58**	61.45±2.65
4	36.59±11.92	82.33±12.97	78.36±9.35	45.68±14.33	65.78±9.75	94.26±5.71	78.19±15.15	88.10±1.38	90.31±7.68	**97.50±3.15**	83.97±0.84
5	62.22±12.31	78.28±7.00	73.08±6.40	71.93±18.86	64.81±15.74	80.92±7.28	73.93±7.51	71.17±0.72	**81.82±6.17**	80.77±7.20	79.01±1.35
6	62.99±12.76	92.63±7.38	82.46±6.53	89.82±7.34	73.86±5.07	93.57±2.79	87.45±4.87	**97.99±0.76**	86.39±3.89	**94.63±3.06**	91.51±5.48
7	87.39±14.87	**100.00±0.00**	**100.00±0.00**	29.73±11.02	99.57±1.30	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
8	86.26±9.50	89.20±14.06	81.44±13.65	98.73±3.19	83.34±5.00	**99.87±0.27**	81.61±15.57	96.83±2.63	98.50±5.73	99.81±0.44	99.37±0.13
9	98.00±4.27	**100.00±0.00**	98.67±2.67	14.41±12.34	98.67±2.67	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	92.50±7.50
10	52.08±13.49	67.34±9.32	59.06±7.05	59.88±16.00	55.53±10.58	71.04±7.72	68.11±5.28	63.47±2.01	**72.77±3.67**	70.82±10.36	57.56±3.24
11	48.90±16.67	51.31±13.58	59.23±12.48	76.15±9.89	48.94±7.60	70.52±12.38	61.44±11.39	69.02±2.02	73.09±8.80	73.47±9.50	**85.48±1.41**
12	27.57±6.28	60.20±16.55	44.15±10.39	48.77±15.10	25.43±9.98	69.06±10.32	58.78±15.39	82.79±1.50	56.09±9.89	76.90±20.20	**94.86±0.08**
13	91.35±11.30	99.15±1.18	99.10±0.97	90.68±6.00	89.80±6.89	97.85±2.96	98.90±1.11	99.00±0.71	99.49±1.72	98.90±1.30	**100.00±0.00**
14	62.41±14.60	84.63±9.27	81.48±12.04	90.16±8.30	89.37±4.66	94.84±7.10	84.54±16.45	81.32±5.27	90.92±6.21	93.40±8.97	**97.23±2.13**
15	47.56±8.95	73.86±15.27	70.37±12.97	65.91±13.09	58.74±9.69	88.50±10.89	74.02±16.86	**99.90±0.13**	85.37±12.25	86.27±10.47	86.92±9.97
16	93.64±4.34	98.52±1.90	98.64±2.58	63.45±26.63	97.05±4.85	97.04±3.77	98.86±1.02	94.32±3.05	100.00±0.00	99.77±0.45	98.92±0.15
OA (%)	50.55±2.68	67.02±3.04	62.99±2.94	67.39±3.86	55.81±1.79	77.02±3.55	70.57±3.47	72.24±0.67	75.75±2.65	79.07±3.39	**82.71±1.02**
AA (%)	62.66±1.56	80.40±1.61	75.59±1.11	64.03±3.18	68.82±0.98	86.25±1.53	80.62±1.91	84.34±0.54	84.63±1.68	**87.78±1.90**	87.68±2.63
Kappa	44.60±2.65	63.04±3.21	58.37±3.01	63.58±4.13	50.58±1.80	74.19±3.88	66.88±3.63	68.99±0.72	72.54±2.93	76.38±3.74	**80.46±1.67**

E. Generalization Performance

To thoroughly assess the generalization capability of KDMSC, we conduct a comparative study with other classification methods under the constraint of limited training data. Specifically, on four benchmark datasets, we randomly sample a fixed number of labeled instances per class (i.e., 5, 10, 15, 20, and 25) and depict the variation of OA with respect to the number of labeled samples, as illustrated in Fig. 8. The results reveal that KDMSC consistently attains the highest classification accuracy on most datasets, clearly validating the strong generalization ability of the proposed approach.

F. Classification Results

1) Results on the Indian Pines: According to the results in Table III, it can be observed that KDMSC outperforms other comparative methods, achieving the highest OA. Specifically, our method achieves an accuracy of 82.71% ± 1.02%, representing a 3.64% improvement over the best-performing baseline. Furthermore, KDMSC shows the highest classification accuracy in five categories: 7, 11, 12, 13, and 14. However, the performance across all methods remains suboptimal for category 3. In this category, KDMSC achieves an accuracy of 61.45% ± 2.65%, which is superior to baselines such as 3DCNN and RN-FSC, but still lags behind SPFormer by 9.8%. We identify two primary factors contributing to the performance gap in category 3. First, category 3 exhibits significant spectral overlap with adjacent classes and a relatively sparse sample distribution, resulting in unclear classification boundaries. Second, capturing the fine-grained local features of these complex samples presents a challenge for the current model. Nonetheless, KDMSC demonstrates remarkable overall performance, achieving high AA and a Kappa coefficient, highlighting its robustness and reliability.

Fig. 9 presents a visual comparison of the classification maps generated by different methods on the Indian Pines dataset. Upon observation, the classification map generated by KDMSC (see Fig. 9) closely aligns with the ground-truth map, thereby validating the effectiveness and reliability of our method.

2) Results on the University of Pavia: According to the results presented in Table IV, the proposed KDMSC method demonstrates significant advantages over other methods in terms of performance, achieving the highest accuracy among all comparison methods. Specifically, KDMSC achieves an accuracy of 92.60% ± 0.99%, which is an improvement of 4.05% over the best baseline method, MLPA. In terms of accuracy across categories, KDMSC performs best in classes 3 and 8, with a particularly notable improvement of 7.45% in class 8 compared with the second-best method, SPFormer. This advantage stems from the dual-perspective graph strategy in KDMSC, which helps the model to capture finer features, thereby enhancing its classification capability. Moreover, KDMSC also shows significant improvements in evaluation metrics such as AA and Kappa coefficients, further confirming the superiority of KDMSC.

Fig. 10 provides a visual comparison of the classification maps generated by different methods on the University of Pavia dataset. The classification maps obtained by other comparison methods exhibit errors in several regions. In contrast,

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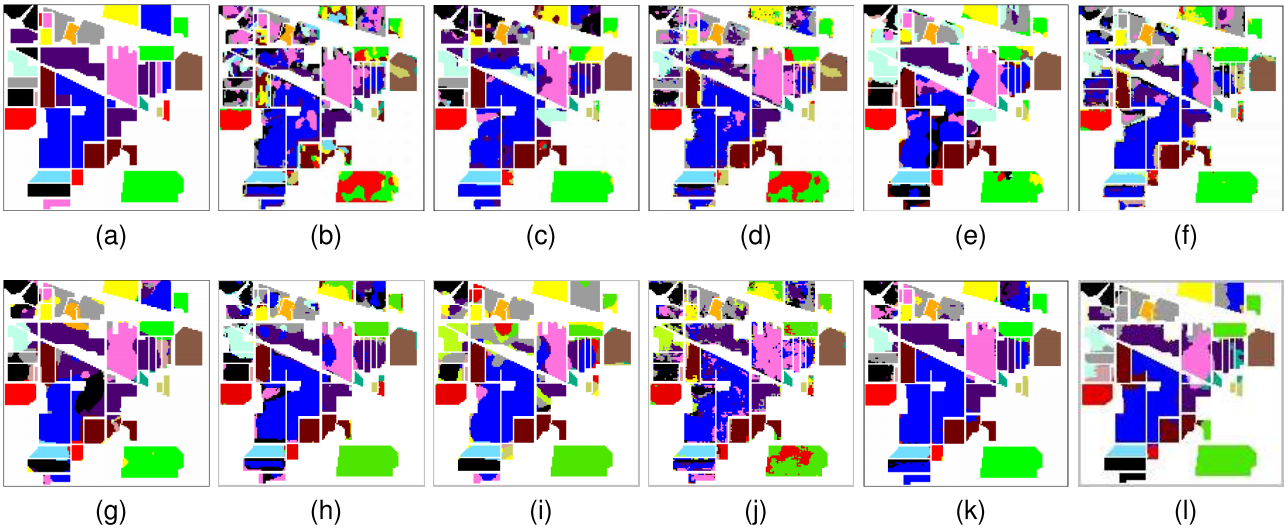


Fig. 9. Classification maps of different methods for the Indian Pines. (a) Ground truth. (b) 3DCNN. (c) SSRN. (d) DCFSL. (e) DPN-HRA. (f) RN-FSC. (g) S3Net. (h) HyMuT. (i) HSI-MFormer. (j) MLPA. (k) SPFormer. (l) KDMSC.

TABLE IV
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR UNIVERSITY OF PAVIA

Class	3DCNN [10]	SSRN [9]	DCFSL [39]	DPN-HRA [48]	RN-FSC [55]	S3Net [71]	HyMuT [45]	HSI-MFormer [16]	MLPA [42]	SPFormer [15]	KDMSC
1	55.01±11.49	85.71±5.50	77.69±12.85	**91.84±5.35**	72.76±12.97	85.99±7.48	80.78±14.12	74.31±1.02	82.33±5.18	89.69±5.57	86.73±1.19
2	68.25±9.77	61.21±17.80	87.38±3.86	94.54±4.09	83.44±4.30	81.62±14.35	87.56±7.55	90.48±2.76	**97.77±8.88**	75.70±10.46	93.32±1.78
3	61.47±8.17	65.72±14.23	65.53±12.97	44.87±12.66	67.36±11.15	82.82±7.02	70.36±8.01	73.88±2.25	69.55±7.31	83.21±9.43	**95.97±0.17**
4	77.45±5.35	91.22±5.27	92.50±4.36	63.97±16.20	91.00±3.89	89.41±4.48	92.63±3.18	68.66±1.69	**93.61±3.22**	91.27±4.96	85.70±0.78
5	83.71±10.47	96.87±5.46	99.31±0.98	98.29±1.90	99.44±0.53	99.27±1.07	98.63±1.03	**99.97±0.08**	99.55±0.59	99.93±0.20	99.63±0.37
6	48.58±5.71	77.10±13.20	74.07±5.47	65.75±21.53	57.86±5.41	84.79±10.31	73.39±11.85	77.89±4.57	64.67±9.60	**93.63±8.76**	93.23±3.81
7	78.40±6.99	93.21±6.04	78.94±7.90	42.61±23.60	68.94±8.79	**99.77±0.40**	75.73±13.03	98.75±0.81	80.83±5.19	99.65±0.44	98.95±0.15
8	60.33±11.98	81.56±8.79	63.53±11.79	58.27±12.19	72.20±10.04	84.29±18.06	73.77±16.93	35.52±4.95	88.62±10.41	89.78±6.65	**97.23±1.85**
9	87.77±12.16	**99.71±0.60**	99.11±0.76	79.93±15.11	98.45±1.64	82.24±10.76	92.63±9.37	86.77±3.04	97.65±2.66	95.61±5.89	90.29±6.86
OA (%)	64.76±3.95	73.96±6.82	81.86±1.91	70.27±5.23	77.95±2.05	84.73±5.57	83.27±0.98	79.86±1.78	88.55±3.85	84.61±4.36	**92.60±0.99**
AA (%)	69.00±2.66	83.59±2.82	81.81±1.47	71.12±3.09	79.05±1.00	87.82±2.95	82.83±1.74	78.47±1.16	86.07±2.18	90.94±1.66	**93.47±1.88**
Kappa	55.50±4.17	67.89±7.31	76.41±2.31	62.82±5.90	71.40±2.38	80.56±6.45	78.15±0.17	73.65±2.23	84.62±4.65	80.67±5.07	**90.31±1.27**

TABLE V
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR SALINAS

Class	3DCNN [10]	SSRN [9]	DCFSL [39]	DPN-HRA [48]	RN-FSC [55]	S3Net [71]	HyMuT [45]	HSI-MFormer [16]	MLPA [42]	SPFormer [15]	KDMSC
1	94.85±6.05	99.57±0.70	98.99±1.12	99.52±0.87	96.26±2.47	98.40±2.98	98.84±1.82	**100.00±0.00**	98.30±1.46	99.58±1.21	**100.00±0.00**
2	95.17±7.18	99.61±0.49	99.59±0.92	98.59±1.30	98.71±1.09	97.13±5.71	99.55±0.42	99.89±0.23	99.22±0.48	**100.00±0.00**	99.89±0.19
3	89.56±5.79	89.10±6.81	90.44±10.53	96.11±5.14	82.51±5.25	97.98±5.08	98.13±1.00	**100.00±0.00**	99.59±2.81	94.52±7.73	99.16±1.42
4	98.91±0.66	99.45±1.08	99.48±0.62	89.54±6.68	98.77±1.28	98.91±2.08	98.99±0.57	99.70±0.32	97.83±0.82	**99.94±0.08**	99.61±0.48
5	93.98±2.92	91.11±10.80	91.08±3.03	95.81±3.16	90.63±6.78	97.36±2.74	92.59±0.82	94.83±1.07	94.79±2.64	**97.65±1.86**	96.58±1.62
6	98.58±1.89	99.57±0.68	98.44±2.20	99.32±0.68	98.73±0.97	99.10±1.05	99.33±0.86	99.68±0.33	**100.00±0.00**	99.48±0.79	99.64±0.67
7	96.72±2.93	99.80±0.25	98.98±1.77	98.41±2.57	93.85±4.49	99.92±0.20	99.14±0.59	**99.99±0.01**	99.92±0.29	99.94±0.07	97.90±1.75
8	60.13±11.11	66.02±15.88	77.86±9.76	92.73±4.12	68.69±9.68	80.37±9.15	80.85±1.24	76.17±2.45	80.45±6.11	83.03±6.67	**94.49±3.05**
9	94.49±2.67	96.92±3.07	99.07±1.26	99.00±0.92	91.43±1.46	99.45±1.19	98.64±1.07	99.99±0.02	98.79±0.60	**100.00±0.00**	99.73±0.47
10	76.22±11.69	84.76±6.13	87.02±3.49	94.39±4.45	71.12±5.56	91.35±5.73	87.13±8.84	**95.04±1.09**	93.76±5.52	93.71±4.83	92.51±2.31
11	87.63±7.32	96.99±5.52	98.57±1.36	96.47±5.06	82.81±15.07	99.32±1.04	98.53±0.69	**100.00±0.00**	98.20±1.05	99.94±0.12	97.73±3.56
12	97.64±3.42	98.55±2.55	99.61±0.56	91.79±6.14	93.13±3.48	91.10±11.22	99.29±0.86	95.34±2.51	**98.90±0.71**	97.10±3.39	93.71±9.66
13	98.12±2.25	99.20±1.22	99.25±0.34	86.33±18.63	96.02±2.72	97.52±3.40	**99.26±0.61**	92.65±2.81	98.12±0.65	98.10±1.34	98.55±0.88
14	96.94±2.46	97.90±1.68	98.55±1.73	88.56±16.30	97.91±3.27	94.74±6.07	98.81±0.23	93.16±1.54	99.72±0.65	**99.85±0.21**	97.99±0.42
15	72.09±7.97	75.89±7.83	73.04±6.60	78.56±6.25	68.20±6.21	90.65±14.73	77.33±9.34	90.69±2.05	83.54±7.19	86.92±9.19	**92.62±4.34**
16	81.18±12.00	85.02±7.20	91.71±6.05	96.68±7.07	82.64±10.00	95.70±4.61	94.36±3.32	97.07±1.96	92.93±5.67	97.95±2.48	**98.20±0.24**
OA (%)	83.27±2.32	56.80±2.82	89.46±1.68	92.69±1.10	83.34±1.75	92.89±1.19	91.11±1.06	92.66±0.71	92.39±1.50	93.74±1.18	**96.60±0.39**
AA (%)	89.51±1.57	92.47±1.54	93.86±0.76	93.86±2.26	88.21±1.49	95.56±0.91	95.05±0.72	95.89±0.30	95.88±0.80	96.74±0.75	**97.39±1.91**
Kappa	81.46±2.54	85.37±3.08	88.29±1.84	91.88±1.21	81.50±1.93	92.10±1.33	90.12±1.19	91.86±0.78	91.55±1.66	93.05±1.31	**96.22±0.43**

the classification map generated by the proposed KDMSC method presents a smoother visual appearance and shows fewer misclassifications compared with the other methods.

3) Results on the Salinas: According to the results in Table V, nearly all methods achieve good classification performance on this dataset, primarily due to the more regular land boundary patterns in this dataset compared with others. Among them, the KDMSC method significantly outperforms the other comparison methods, achieving the highest overall classification accuracy (96.60% ± 0.39%), with an improvement of 2.86% over the optimal baseline method, SPFormer. In terms of class accuracy, KDMSC has achieved suboptimal or optimal classification performance for most land cover classes, particularly for class 8, where KDMSC has shown a significant improvement of 11.46% over SPFormer. In addition, KDMSC also demonstrates notable improvements in the AA and Kappa coefficients compared with other methods. These results further confirm the superiority of KDMSC.

Fig. 11 presents a visual comparison of the classification maps generated by different methods on the Salinas dataset. It

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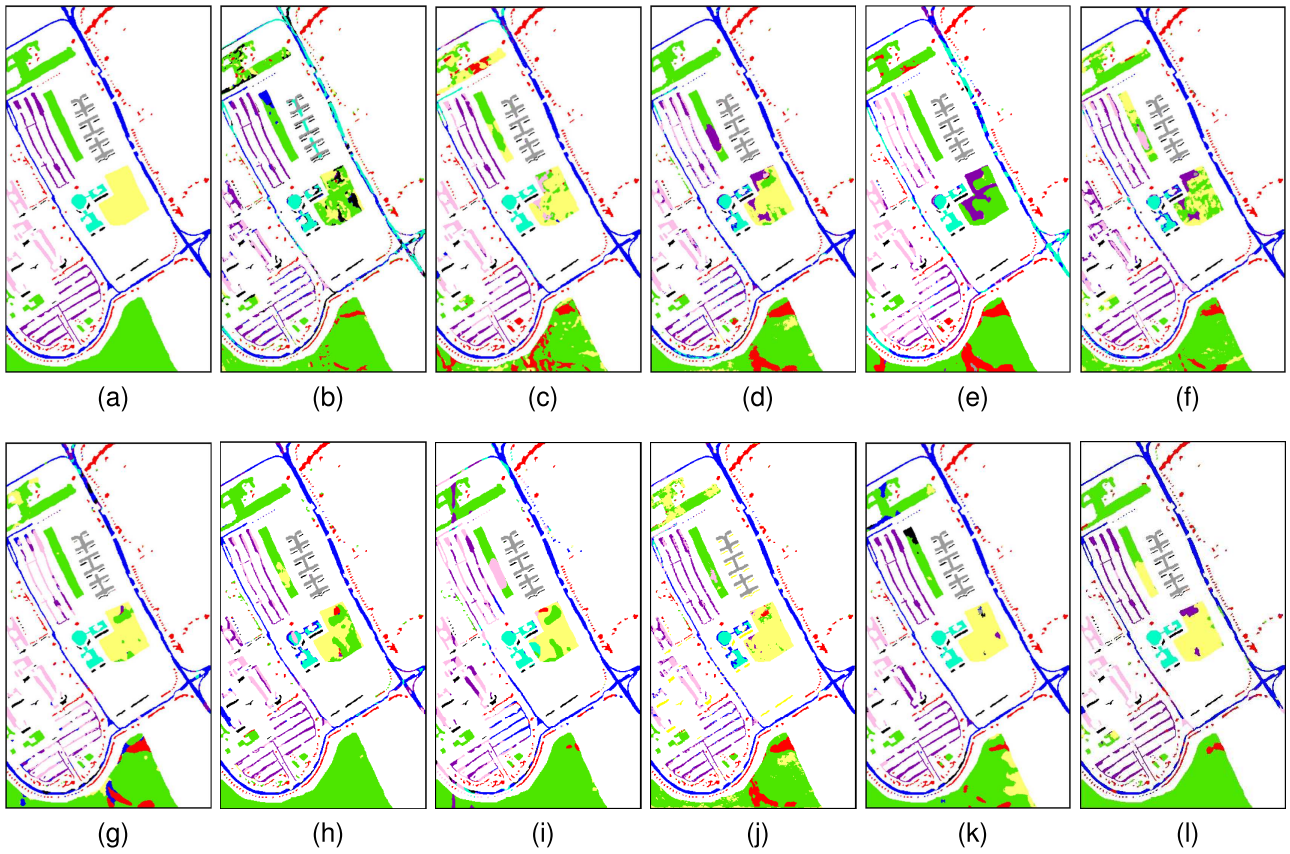


Fig. 10. Classification maps of different methods for the University of Pavia. (a) Ground truth. (b) 3DCNN. (c) SSRN. (d) DCFSL. (e) DPN-HRA. (f) RN-FSC. (g) S3Net. (h) HyMuT. (i) HSI-MFormer. (j) MLPA. (k) SPFormer. (l) KDMSC.

TABLE VI

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR HOUSTON

Class	3DCNN [10]	SSRN [9]	DCFSL [39]	DPN-HRA [48]	RN-FSC [55]	S3Net [71]	HyMuT [45]	HSI-MFormer [16]	MLPA [42]	SPFormer [15]	KDMSC
1	87.78±5.36	86.57±9.17	88.92±6.40	86.52±7.70	85.69±7.33	74.40±11.38	**96.30±3.07**	79.81±1.48	94.52±4.65	85.27±10.59	84.41±2.30
2	74.96±7.59	80.94±7.93	79.03±11.23	77.25±15.13	84.88±6.66	**86.58±6.10**	85.26±4.51	77.69±3.56	83.62±6.29	82.02±9.35	60.42±2.73
3	78.12±37.26	**99.62±0.95**	96.53±3.19	98.67±2.24	86.01±7.12	99.35±0.98	94.15±3.82	91.59±2.92	97.00±2.80	99.58±0.62	95.70±1.92
4	88.32±3.95	92.08±3.53	90.79±3.21	81.81±13.16	**94.96±2.28**	91.34±5.78	93.84±2.23	69.17±2.08	89.11±2.48	92.72±2.89	85.16±3.17
5	93.19±5.05	93.51±9.01	95.24±3.80	89.28±8.74	90.49±11.72	88.24±7.92	97.18±3.13	**99.95±0.10**	97.27±4.45	99.47±0.73	99.60±0.57
6	74.41±5.00	87.91±7.00	80.97±6.58	94.45±5.35	77.94±5.67	90.00±5.80	80.66±7.30	**98.31±0.38**	83.21±4.95	87.69±4.31	91.08±3.92
7	58.99±23.23	75.35±9.10	63.06±13.90	63.49±17.86	72.38±9.24	63.61±8.70	66.08±12.27	58.45±5.54	63.29±9.91	**82.49±12.36**	70.03±5.76
8	35.59±9.80	34.88±9.66	44.67±5.01	**66.60±20.55**	39.59±19.77	41.98±7.86	55.17±10.11	38.97±2.05	58.82±7.57	38.64±8.23	53.05±2.24
9	62.32±8.95	73.24±10.47	63.18±9.36	44.30±13.42	59.05±9.02	51.11±4.47	60.58±5.89	64.09±1.88	69.73±8.26	**78.60±7.99**	78.17±6.67
10	38.89±15.86	61.31±17.88	66.04±14.39	47.61±10.97	63.86±17.38	64.79±18.13	72.39±12.24	71.78±3.17	75.94±9.58	70.23±20.70	**95.71±1.34**
11	57.09±11.22	66.22±9.92	62.93±8.75	62.02±23.41	56.19±9.69	79.18±10.03	63.50±19.90	**84.16±1.94**	69.47±9.80	79.59±7.31	81.24±6.44
12	43.65±8.33	52.93±11.99	73.66±11.54	55.81±15.07	60.78±16.47	50.73±10.76	73.68±18.77	77.92±4.29	80.92±8.89	51.88±10.20	**85.56±1.21**
13	63.58±8.75	90.04±4.29	86.66±7.66	54.87±18.20	**95.60±1.47**	90.91±3.36	84.37±9.80	93.88±0.98	88.50±5.32	84.14±11.35	89.48±3.82
14	89.67±7.18	99.79±0.56	88.30±8.13	95.29±5.21	96.45±3.91	**100.00±0.00**	95.28±4.20	**100.00±0.00**	94.16±3.91	99.95±0.09	98.36±0.57
15	96.44±2.02	**100.00±0.00**	95.73±2.57	85.26±8.17	97.15±2.50	99.85±0.46	97.93±2.08	99.97±0.06	96.95±2.45	**100.00±0.00**	98.33±0.12
OA (%)	67.13±3.50	75.99±2.33	75.91±2.67	67.09±3.58	74.33±2.44	73.93±2.72	78.61±2.21	76.30±0.44	80.80±1.52	79.44±2.13	**81.99±1.75**
AA (%)	69.53±3.35	79.63±1.68	78.38±2.41	73.55±3.53	77.40±1.81	78.14±2.22	81.09±2.19	80.38±0.41	82.83±1.63	82.15±1.99	**84.42±2.85**
Kappa	64.51±3.73	74.11±2.50	73.38±2.88	64.49±3.85	72.28±2.63	71.87±2.93	76.89±2.38	74.42±0.47	79.25±1.64	77.81±2.30	**80.54±1.90**

can be observed that certain regions in the KDMSC classification map (see Fig. 11) exhibit less noise compared with those generated by other methods. This observation corresponds to the results listed in Table V.

4) Results on the Houston: According to the results in Table VI, KDMSC exhibits the best overall performance among all compared methods, achieving the highest overall classification accuracy. Specifically, KDMSC attains an accuracy of 81.99 ± 1.75%, surpassing other methods by a margin of 1.19%–14.9%. In terms of per-class accuracy, KDMSC achieves the best results in categories 10 and 12, significantly outperforming other approaches. Notably, in category 10, KDMSC exceeds the second-best method, MLPA, by 19.77%. In contrast, KDMSC underperforms S3Net in category 2, primarily due to the increased spectral complexity within this class, which introduces challenges for stable and effective feature extraction. While KDMSC demonstrates clear advantages in OA, its capacity to model intricate class-specific features in such complex cases remains limited. S3Net, on the other hand, likely incorporates more refined strategies tailored for complex sample structures, thereby achieving superior performance in this category. Nevertheless, KDMSC surpasses all other methods in terms of AA and Kappa coefficient, further confirming its superiority in terms of overall performance and robustness.

Fig. 12 presents a comparative visualization of the classification maps produced by various methods on the Houston dataset. It is evident that the proposed method delivers

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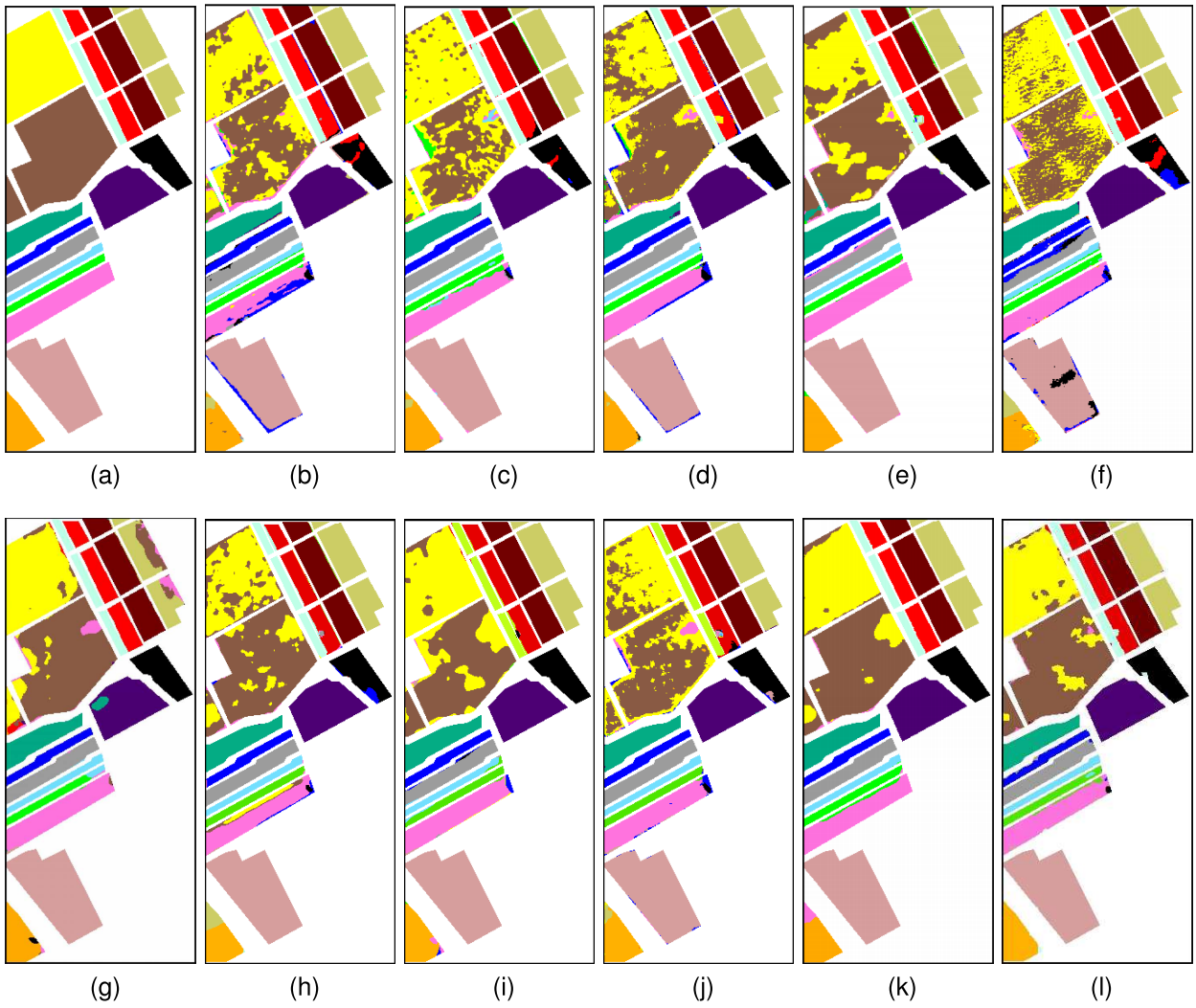


Fig. 11. Classification maps of different methods for the Salinas. (a) Ground truth. (b) 3DCNN. (c) SSRN. (d) DCFSL. (e) DPN-HRA. (f) RN-FSC. (g) S3Net. (h) HyMuT. (i) HSI-MFormer. (j) MLPA. (k) SPFormer. (l) KDMSC.

superior visual results, demonstrating overall better classification performance.

V. Discussion

A. Feature Combination Analysis in DSGKD

In DSGKD, we extract three types of superpixel features: $\mathbf{H}_s$, $\mathbf{H}_u$, and $\mathbf{H}_t$. After completing the knowledge distillation, to further enhance feature representation, we employ AWF to first perform feature-weighted fusion of the two superpixel features. To explore the performance of different superpixel feature combinations across different datasets, we compare three variants of KDMSC: KDMSCut, KDMSCst, and KDMSCsu. Specifically, KDMSCut and KDMSCst refer to the combinations of student network features and teacher network features, while KDMSCsu uses only the student network features for fusion.

Based on the experimental findings presented in Table VII, the performance disparities among the three variants are negligible. This can be attributed to the DSGKD framework's mechanism, where the student and teacher networks engage in a mutual learning process: the teacher model's predictions for all samples are distilled back to the student model, and conversely, the advancements in the student model contribute to the further optimization of the teacher model's performance. Specifically, KDMSCsu excels on the Pavia, Salinas, and Houston datasets, whereas KDMSCst demonstrates a slightly superior accuracy on the Indian Pines dataset. These results suggest that, depending on the scenario, the strategic selection of feature combinations can significantly bolster the model's performance.

TABLE VII
Comparison of Classification Performance With Different Feature Combinations

	Indian Pines	Pavia	Salinas	Houston
KDMSCut	81.89 ± 1.64	92.43 ± 1.35	96.36 ± 1.85	81.25 ± 0.81
KDMSCst	82.71 ± 1.02	92.26 ± 1.61	96.47 ± 0.73	81.63 ± 0.97
KDMSCsu	82.67 ± 1.49	92.60 ± 0.99	96.60 ± 0.39	81.99 ± 1.75

B. Feature Fusion Visualization

To further analyze the effectiveness of feature fusion in the proposed KDMSC framework, the visualization results

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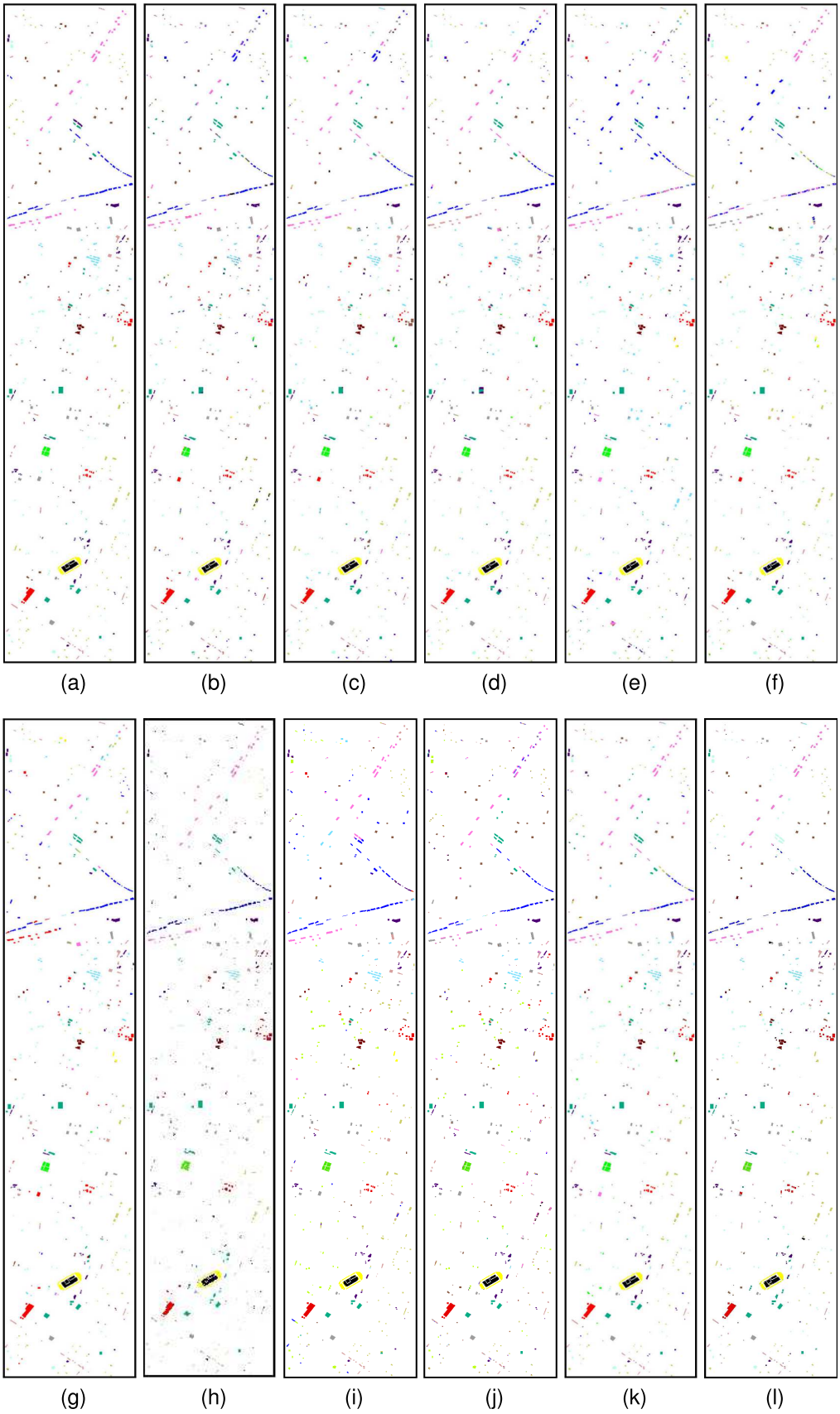


Fig. 12. Classification maps of different methods for the Houston. (a) Ground truth. (b) 3DCNN. (c) SSRN. (d) DCFSL. (e) DPN-HRA. (f) RN-FSC. (g) S3Net. (h) HyMuT. (i) HSI-MFormer. (j) MLPA. (k) SPFormer. (l) KDMSC.

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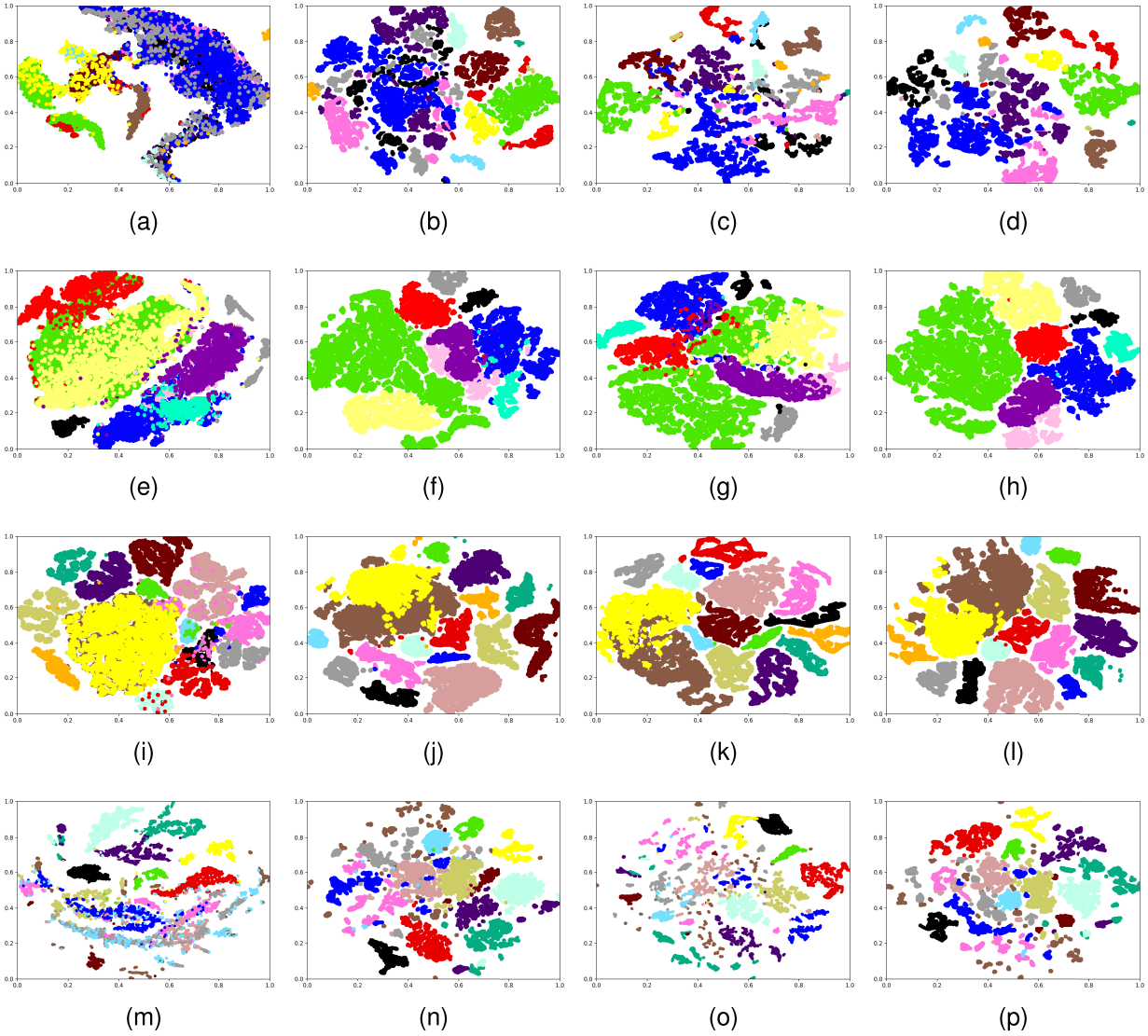


Fig. 13. t-SNE-based visualization of feature distributions for different methods. Raw features for (a) Indian Pines, (e) University of Pavia, (i) Salinas, and (m) Houston. MLPA for (b) Indian Pines, (f) University of Pavia, (j) Salinas, and (n) Houston. SPFormer for (c) Indian Pines, (g) University of Pavia, (k) Salinas, and (o) Houston. KDMSC for (d) Indian Pines, (h) University of Pavia, (l) Salinas, and (p) Houston.

of local pixel and global superpixel features, as well as their fusion outcomes, are presented in the Supplementary Material (see Section II). These visualizations demonstrate the complementary advantages of the two feature branches and verify the enhanced spatial consistency achieved by the fusion mechanism.

C. t-SNE Feature Visualization

To provide a more intuitive analysis of the discriminative capability of the features learned by the KDMSC method, we conduct a t-SNE-based visualization on the original data, the strongest baseline methods (MLPA and SPFormer), and the high-dimensional features extracted by KDMSC, with results reported on four benchmark datasets (see Fig. 13).

The visualization results clearly demonstrate that the features extracted by KDMSC exhibit significantly better intraclass compactness and interclass separability than those of MLPA and SPFormer. Specifically, KDMSC forms more compact clusters for the majority of categories, while simultaneously producing sharper boundaries between different categories, thereby showing stronger discriminative power. This superiority stems from the overall design of KDMSC: at the superpixel level, DSGKD module effectively integrates graph-structural information from multiple perspectives, enhancing the representation of global spatial features; at the pixel level, the MSFC module captures fine-grained spectral and spatial features across multiple receptive fields; meanwhile, the AWF module serves as a dynamic bridge during fusion, balancing the contributions of pixel-level and superpixel-level features so that the final representation preserves local details while maintaining global consistency. Consequently, the KDMSC method is able to learn more discriminative feature representations in the embedding space, thereby providing a more reliable foundation for subsequent classification tasks.

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TABLE VIII
RUNNING TIME, FLOPS, PARAMETER SIZE, AND PEAK MEMORY OF DIFFERENT METHODS

Dataset	Metric	DCFSL [39]	S3Net [71]	HyMuT [45]	HSI-MFormer [16]	MLPA [42]	SPFormer [15]	KDMSC
Indian Pines	Parameters	4288996	164464	4690000	132200	12660000	86296	824396
	Train(s)	3821.83	5.84	4548.17	114.23	2202.40	18.92	163.21
	Test(s)	29.44	6.49	0.93	1.62	1.01	3.08	7.99
	FLOPs(G)	0.09	0.65	0.10	0.07	4.13	0.02	3.29
	Peak Memory(MB)	1246.38	306.92	2485.96	736.29	835.46	663.50	944.63
Paviau	Parameters	4259294	100973	4690000	132200	12650000	45225	468623
	Train(s)	3568.57	5.23	5282.67	93.26	1849.13	326.64	952.89
	Test(s)	31.93	20.23	3.16	7.15	7.65	8.07	10.59
	FLOPs(G)	0.09	0.34	0.10	0.07	3.48	0.01	4.66
	Peak Memory(MB)	824.39	124.08	2475.03	723.52	782.51	185.34	875.08
Salinas	Parameters	4269394	164464	4690000	132200	12650000	86456	89401
	Train(s)	378.48	16.91	4595.35	139.89	2018.93	50.48	1497.62
	Test(s)	33.21	28.44	4.96	8.90	2.93	16.43	37.94
	FLOPs(G)	0.09	0.65	0.10	0.07	3.48	0.02	5.83
	Peak Memory(MB)	1179.63	341.00	2493.55	917.26	822.34	663.72	903.09
Houston	Parameters	4264360	164464	4680000	132200	12650000	46735	243502
	Train(s)	3683.13	22.49	4244.35	152.63	3344.71	461.38	1508.36
	Test(s)	30.13	37.09	5.23	11.29	2.59	1.70	59.32
	FLOPs(G)	0.09	0.65	0.10	0.07	3.63	0.02	5.62
	Peak Memory(MB)	1208.06	353.20	2498.16	847.77	840.13	663.72	843.02

D. Complexity Analysis

In terms of computational complexity, we select six representative baseline methods for comparison and conduct a comprehensive evaluation on four datasets. As shown in Table VIII, KDMSC demonstrates a well-balanced tradeoff between model size and computational efficiency. Regarding the number of parameters, KDMSC maintains a model size ranging from 89k to 824k, which is significantly smaller than large-scale models such as DCFSL and HyMuT (4.2M–4.6M), but slightly higher than lightweight methods like SPFormer (45k–86k) and S3Net (164k). Compared with recent high-cost baselines such as MLPA and HSI-SPFormer, KDMSC achieves lower resource usage while maintaining a reasonable parameter scale, demonstrating better deployment adaptability. In terms of computational efficiency, KDMSC achieves moderate training and testing times—slightly higher than SPFormer and S3Net, but significantly better than HyMuT and the largest model, MLPA. Its FLOPs range from 3.29G to 5.83G, and peak memory usage remains within 843–944 MB, which is comparable to that of MLPA, demonstrating efficient resource utilization.

The computational overhead of KDMSC mainly comes from two aspects: the MSFC module at the pixel level and the DSGKD module at the superpixel level. In addition, a majority voting-based ensemble strategy is introduced to fuse graph segmentation results across multiple scales, enhancing spatial structural representation. Although these components introduce some computational burden, the overall resource consumption remains within a controllable range while effectively boosting model performance. Specifically, on the University of Pavia dataset, KDMSC takes 952.89 s for training and 10.59 s for testing, which is higher than SPFormer (326.64 and 8.07 s), yet achieves an OA that is 7.99% higher. Importantly, KDMSC achieves FLOPs and memory usage comparable to those of MLPA, and its peak memory consumption is only about one-third of HyMuT's. Such a design strikes a favorable balance between computational cost and classification accuracy, avoiding excessive resource demands while ensuring strong performance. Experimental results show that KDMSC attains significant gains in classification accuracy at the cost of only a moderate increase in computational resources, which highlights its practical value in real-world applications.

VI. CONCLUSION

This article proposes a novel KDMSC method to address the challenges of limited labeled samples and incomplete feature representation in HSI classification. The method integrates a DSGKD module and a multiscale CNN module to extract global and local features from both superpixel and pixel perspectives, respectively. These features are seamlessly integrated using an adaptive weighted fusion strategy. Experimental results show that with just five labeled samples per class, the proposed method outperforms existing deep learning approaches in terms of classification accuracy and generalization performance across four benchmark datasets, significantly improving classification outcomes in few-shot scenarios.

In future work, we intend to incorporate hypergraphs and hypergraph neural networks (HGNNs) to better capture higher order relationships and complex structural information within HSIs. Unlike traditional GNNs, hypergraphs offer flexible modeling of relationships among multiple pixels or superpixels, thus enhancing feature representation. Moreover, to further improve performance on challenging categories, a sample-adaptive weighting strategy should also be designed in future work to enhance the model's ability to learn from difficult samples and boost overall effectiveness. In addition, we also acknowledge the importance of evaluating cross-domain generalization capability and plan to explore the robustness of our approach under domain-shift scenarios in future studies.

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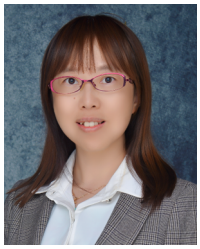
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